

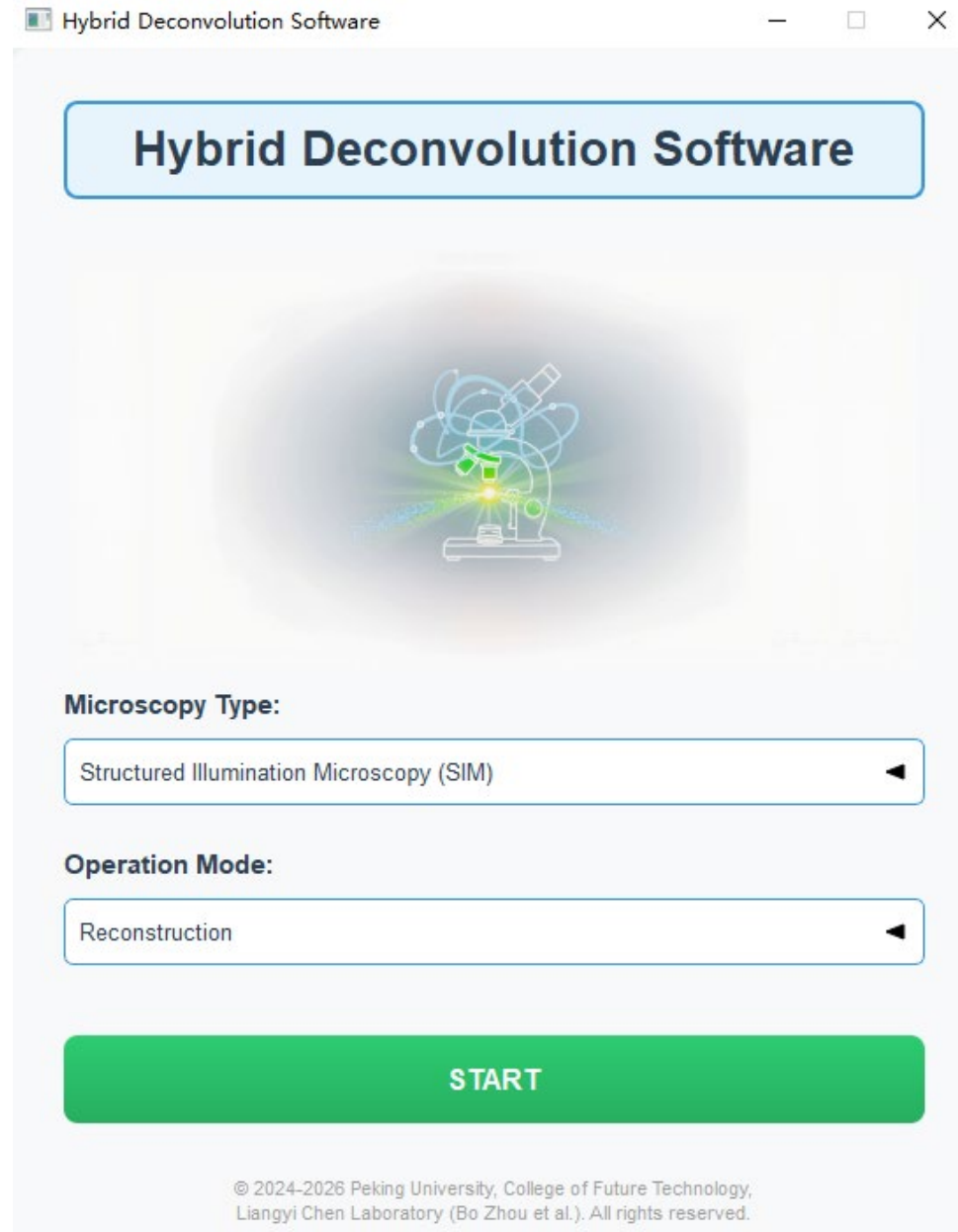
Hybrid Deconvolution Software Handbook

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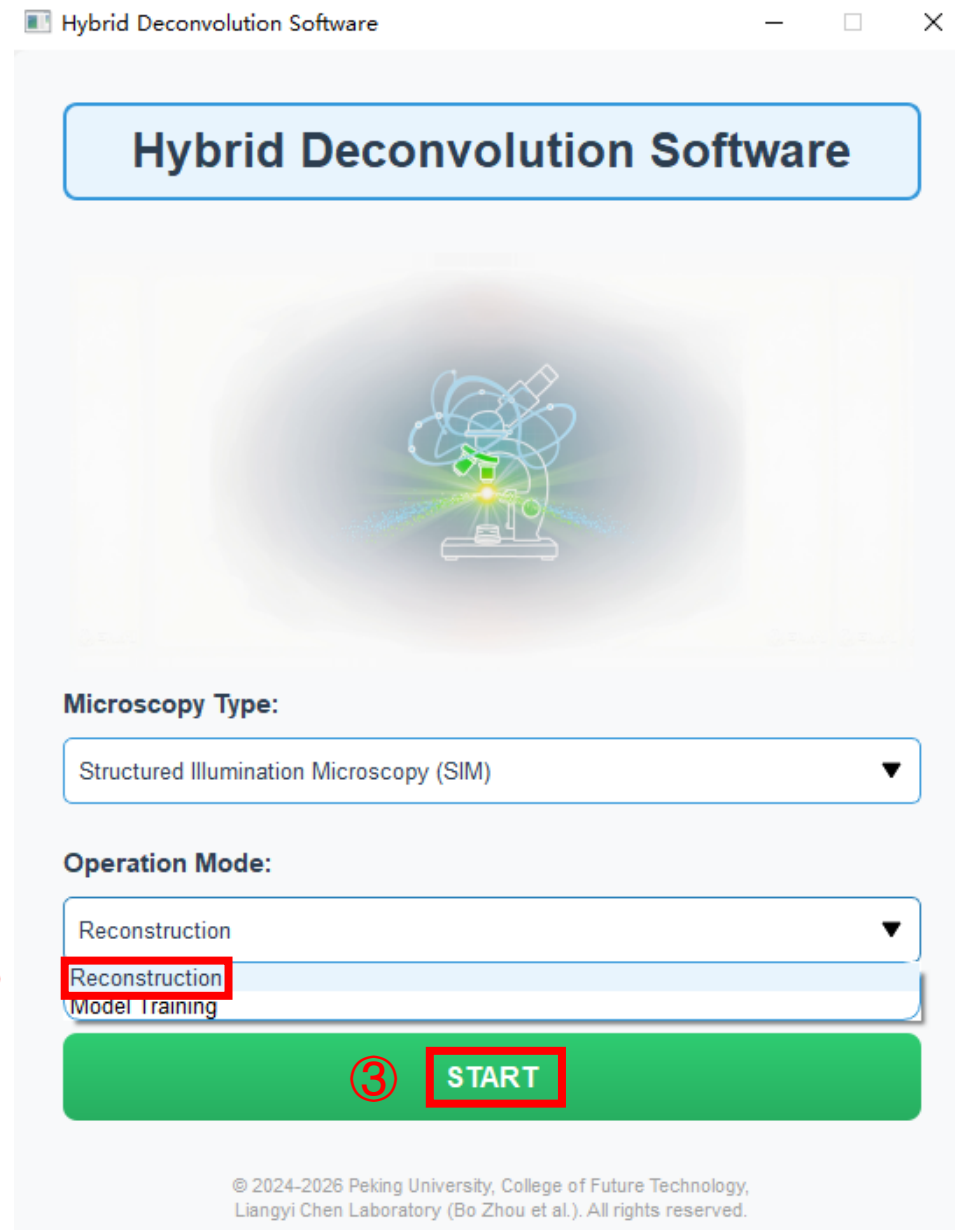
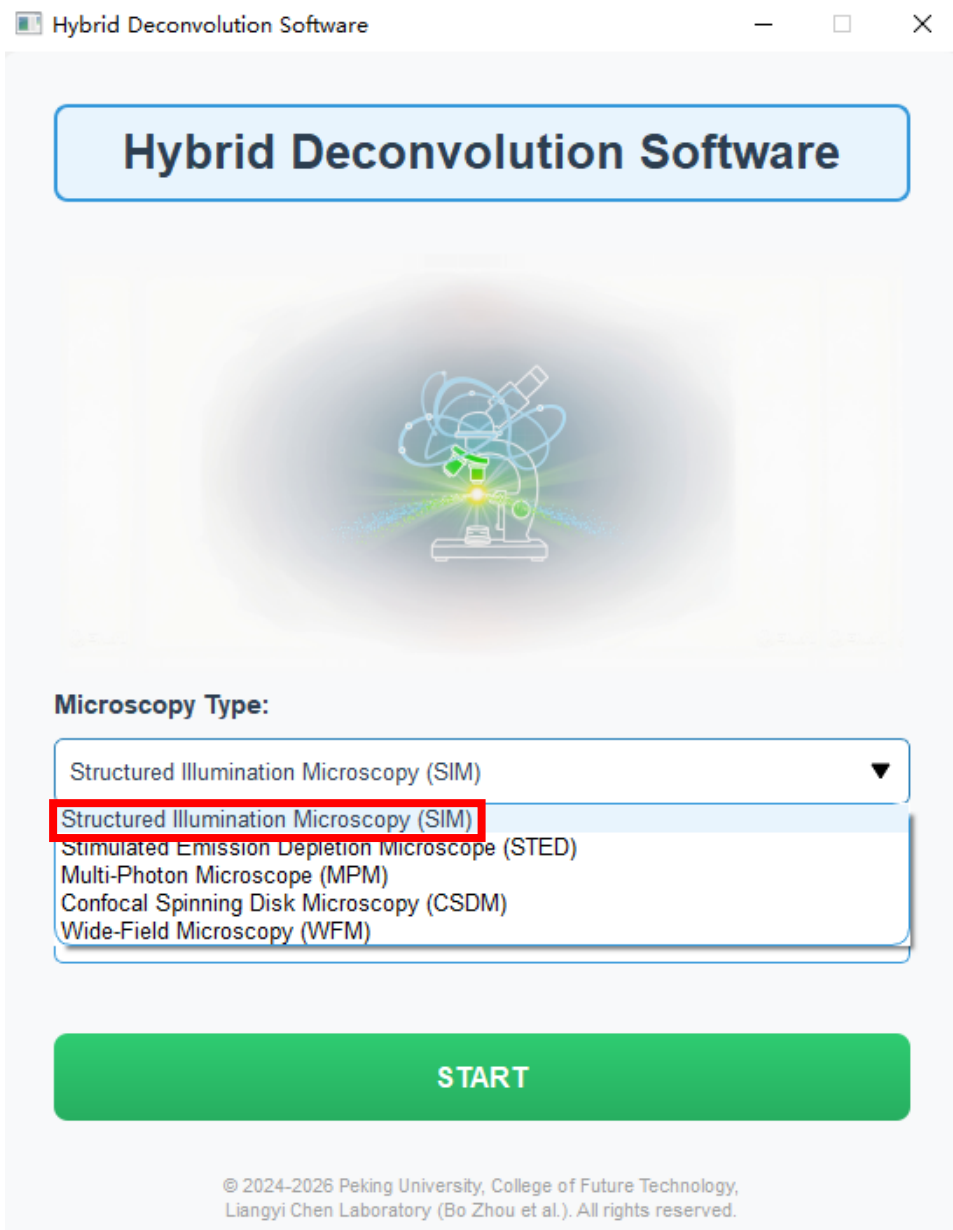
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Hybrid Deconvolution Software Main Graphical User Interface (GUI)



Hybrid-SIM Reconstruction

Start Hybrid-SIM Reconstruction GUI



Hybrid-SIM Reconstruction GUI Layout

① Select parameters

② Run

③ Wait for progress

④ Visualize and export results

The screenshot displays the 'Hybrid-SIM Reconstruction' GUI. The interface is divided into two main panels: 'Hybrid-SIM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right. The left panel features tabs for 'BF-SIM', 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'BF-SIM Parameters' section includes input fields for 'SIM Raw Data Path' (with a 'Browse' button), 'SIM Excitation NA' (1.40), 'SIM Emission NA' (1.40), 'SIM Excitation Wavelength (nm)' (488.00), 'SIM Emission Wavelength (nm)' (525.00), 'SIM Raw Pixel Size (nm)' (43.33), and 'SIM Recon OTF Path' (Simulation). Below these are buttons for 'Default Parameters', 'Load Parameters', and 'Export Parameters'. A 'Run Reconstruction' button is highlighted with a red box. Below it is a 'Stop Reconstruction' button. At the bottom, a green progress bar shows '100%' completion, with the text 'Reconstruction completed!' below it. The right panel, titled 'Reconstruction Results', shows four sub-images: 'BF-SIM Reconstruction', 'BF+Hessian-SIM Reconstruction', 'BF+Hessian+TDV-SIM Reconstruction', and 'Hybrid-SIM Reconstruction'. At the bottom of this panel are sliders for 'Display Range' and 'Frame' (7/10), and buttons for 'Reset Range', 'Reset View', and 'Export Results'. Red arrows point from the numbered annotations to the corresponding GUI elements: ① points to the parameter input fields, ② points to the 'Run Reconstruction' button, ③ points to the progress bar, and ④ points to the reconstruction result images.

Hybrid-SIM Reconstruction

Hybrid-SIM Reconstruction Parameters

BF-SIM Hessian Denoise TDV Denoise Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Path no_Recon_SIM.tif Browse Reset

SIM Excitation NA 1.40 Reset

SIM Emission NA 1.40 Reset

SIM Excitation Wavelength (nm) 488.00 Reset

SIM Emission Wavelength (nm) 525.00 Reset

SIM Raw Pixel Size (nm) 43.33 Reset

SIM Recon OTF Path Simulation Browse Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters Load Parameters Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed!

Reconstruction Results

BF-SIM Reconstruction

BF+Hessian-SIM Reconstruction

BF+Hessian+TDV-SIM Reconstruction

Hybrid-SIM Reconstruction

Display Range: Reset Range Reset View Export Results

Frame: 7/10

Hybrid-SIM Reconstruction Parameters ---- BF-SIM Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Path	no_Recon_SIM.tif	Browse	Reset
SIM Excitation NA	1.40	Reset	
SIM Emission NA	1.40	Reset	
SIM Excitation Wavelength (nm)	488.00	Reset	
SIM Emission Wavelength (nm)	525.00	Reset	
SIM Raw Pixel Size (nm)	43.33	Reset	
SIM Recon OTF Path	Simulation	Browse	Reset

#: Please prepare the raw data TIFF file in advance.

SIM Raw Data Path:

1. TIFF file path of the SIM raw data#;
2. Please choose the file via the Browse button;
3. Sample file: [Demo_Recon_SIM.tif].

SIM Excitation/Emission NA:

1. Excitation/Emission numerical aperture (NA) of the SIM optical system;
2. Please set the values slightly lower than the NA labeled on the objective lens.

SIM Excitation/Emission Wavelength:

1. Excitation/Emission wavelength during the SIM imaging;
2. Please set the values based on the properties of the fluorophore.

SIM Raw Pixel Size:

1. Pixel size of the SIM raw data;
2. Please set the value to the camera pixel's physical size divided by the SIM magnification ratio.

SIM Recon OTF Path:

1. TIFF file path of SIM optical transfer function path;
2. Please select an experimentally measured OTF TIFF file via the Browse button to replace the current default simulated OTF;
3. Sample experimental OTF files are in [src_Hybrid\src_Optics] subfolder within the software installation directory.

Hybrid-SIM Reconstruction Parameters ---- Hessian Denoise Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

Hessian Denoise Parameters

Hessian Fidelity

100.00

Reset

Hessian Z Continuity

2.00

Reset

Hessian Iteration Number

100

Reset

Hessian Fidelity:

1. Fidelity term weight of the Hessian optimization function;
2. Please set the value lower for lower signal-to-noise ratio (SNR) data.

Hessian Z Continuity:

1. The weight for temporal/axial continuity in the Hessian regularization term;
2. Please set the value higher for well-sampled time-lapse or volumetric data.

Hessian Iteration Number:

1. Number of iterations for solving the problem constrained by the Hessian regularization term;
2. Please set the value higher for lower SNR data;
3. Please set the value to 0 for skipping the Hessian Denoise.

Tip: For lower SNR data, please firstly adjust the [Hessian Fidelity], then adjust the [Hessian Iteration Number] if necessary.

Hybrid-SIM Reconstruction Parameters ---- TDV Denoise Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

TDV Denoise Parameters

TDV Model Path

50XSIM_Actin.pth

Browse

Reset

TDV Fidelity

100.00



Reset

TDV Offset

0.10



Reset

TDV Weight

1.00



Reset

TDV Iteration Number

30



Reset

TDV Model Path:

1. PTH file path of the TDV-DNN model;
2. Please choose the file via the Browse button;
3. Please train a new model using the **[Model Training]** mode.

TDV Fidelity:

1. Fidelity term weight of the TDV optimization function;
2. Please set the value lower for lower SNR data.

TDV Offset:

1. Offset value for normalizing the input data;
2. Please set the value higher if apparent signal loss is observed.

TDV Weight:

1. TDV regularization term weight of the TDV optimization function;
2. Please set the value higher for lower SNR data.

TDV Iteration Number:

1. Number of iterations for solving the problem constrained by the TDV regularization term;
2. Please set the value higher for lower SNR data;
3. Please set the value to 0 for skipping the TDV Denoise.

Tip: For low SNR data, please firstly adjust the [TDV Fidelity], then adjust the [TDV Weight] and [TDV Iteration Number] if necessary.

Hybrid-SIM Reconstruction Parameters ---- Sparse Deconvolution Parameters

Hybrid-SIM Reconstruction Parameters

BF-SIM

Hessian Denoise

TDV Denoise

Sparse Deconvolution

Sparse Deconvolution Parameters

Sparse NA

3.00

Reset

Sparse Offset

0.20

Reset

Sparse Iteration Number

25

Reset

Sparse NA:

1. The equivalent NA assumed during point spread function (PSF) generation for sparse deconvolution.
2. Please set the value to ~ 0.8 –2 times the emission NA.

TDV Offset:

1. Offset value for normalizing the input data;
2. Please set the value higher if apparent signal loss is observed.

Sparse Iteration Number:

1. Number of iterations for solving the deconvolution problem constrained by the Sparsity regularization term;
2. Please set the value higher for better resolution improvement;
3. Please set the value to 0 for skipping the Sparse Deconvolution.

Tip: Please begin by setting the [Sparse Iteration Number] (~ 10 –20) and [TDV Offset] (~ 0.1 –0.2). Next, determine the optimal [Sparse NA]. Finally, fine-tune the first two parameters to balance resolution improvement against signal preservation.

Hybrid-SIM Reconstruction Parameters ---- Device

Please select the computing device name (cpu/cuda) for reconstruction.

Hybrid-SIM Reconstruction

Hybrid-SIM Reconstruction Parameters

BF-SIM Hessain Denoise TDV Denoise Sparse Deconvolution

BF-SIM Parameters

SIM Raw Data Pathno_Recon_SIM.tifBrowseReset

SIM Excitation NA1.40Reset

SIM Emission NA1.40Reset

SIM Excitation Wavelength (nm)488.00Reset

SIM Emission Wavelength (nm)525.00Reset

SIM Raw Pixel Size (nm)43.33Reset

SIM Recon OTF PathSimulationBrowseReset

Device:

cpu
cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)
cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Default ParametersLoad ParametersExport Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed !

Reconstruction Results

BF-SIM ReconstructionBF+Hessian-SIM ReconstructionBF+Hessian+TDV-SIM ReconstructionHybrid-SIM Reconstruction

Display Range:Reset RangeReset ViewExport Results

Frame:7/10

Hybrid-SIM Reconstruction Parameters ---- Default, Load and Export Parameters

Please click the button to export the current parameters.

Please click the button to reset the default parameters.

Please click the button to load the previously exported parameters.

The screenshot displays the 'Hybrid-SIM Reconstruction' software interface. The main window is titled 'Hybrid-SIM Reconstruction' and contains two primary sections: 'Hybrid-SIM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-SIM Reconstruction Parameters Section:

- At the top, there are four tabs: 'BF-SIM', 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'BF-SIM' tab is currently selected.
- Below the tabs, the 'BF-SIM Parameters' are listed with input fields and 'Reset' buttons:
 - SIM Raw Data Path: 'no_Recon_SIM.tif' (with 'Browse' and 'Reset' buttons)
 - SIM Excitation NA: '1.40' (with 'Reset' button)
 - SIM Emission NA: '1.40' (with 'Reset' button)
 - SIM Excitation Wavelength (nm): '488.00' (with 'Reset' button)
 - SIM Emission Wavelength (nm): '525.00' (with 'Reset' button)
 - SIM Raw Pixel Size (nm): '43.33' (with 'Reset' button)
 - SIM Recon OTF Path: 'Simulation' (with 'Browse' and 'Reset' buttons)
- Below the parameters, a 'Device:' dropdown menu shows 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' with a note 'Detected: 3 device(s)'.
- At the bottom of this section, three buttons are highlighted with red boxes: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- Below these buttons are two large buttons: 'Run Reconstruction' (green) and 'Stop Reconstruction' (grey).
- A green progress bar at the bottom indicates '100%' completion, with the text 'Reconstruction completed!' below it.

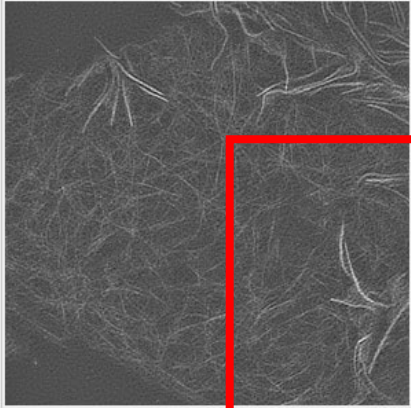
Reconstruction Results Section:

- This section displays four reconstruction results in a 2x2 grid:
 - Top-left: 'BF-SIM Reconstruction' (shows a noisy, low-quality reconstruction of a fiber network).
 - Top-right: 'BF+Hessian-SIM Reconstruction' (shows a clearer, more detailed reconstruction).
 - Bottom-left: 'BF+Hessian+TDV-SIM Reconstruction' (shows a very clear and detailed reconstruction).
 - Bottom-right: 'Hybrid-SIM Reconstruction' (shows a very clear and detailed reconstruction, similar to the bottom-left one).
- Below the grid, there are two sliders for 'Display Range' and 'Reset Range', and two buttons: 'Reset View' and 'Export Results'.
- At the very bottom, a 'Frame:' slider shows the current frame is 7/10.

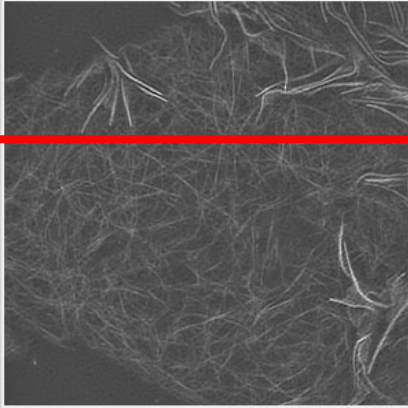
Hybrid-SIM Reconstruction Results Visualization ---- Modify Display Range and Play Video

Reconstruction Results

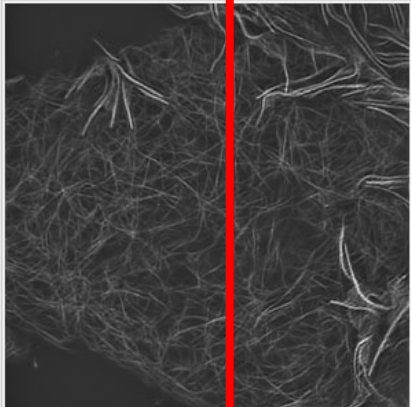
BF-SIM Reconstruction



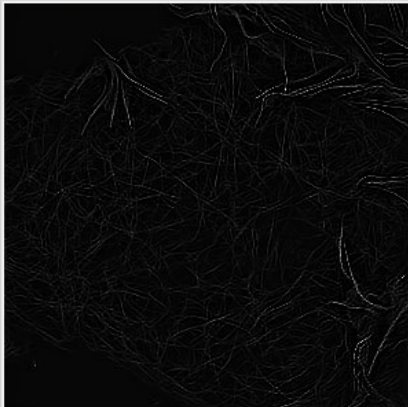
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



Display Range: Reset Range Reset View Export Results

▶ Frame: 1/10

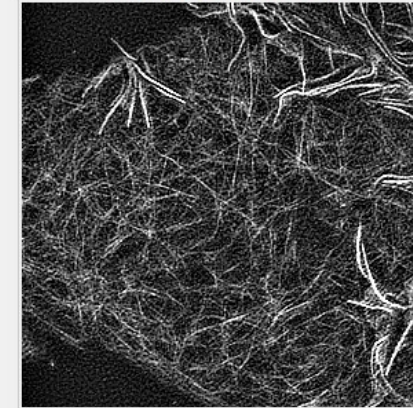
Please use the left and right sliders here to adjust the image display range by setting the minimum and maximum values, respectively.

Please click the button to reset the image display range.

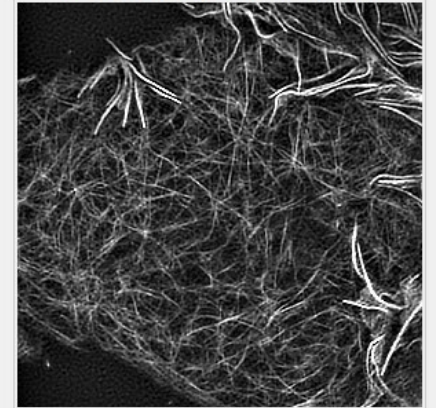
Please use the slider to browse reconstruction results by frame, and click the left button to play the video.

Reconstruction Results

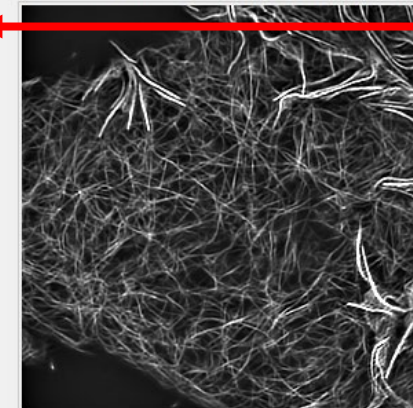
BF-SIM Reconstruction



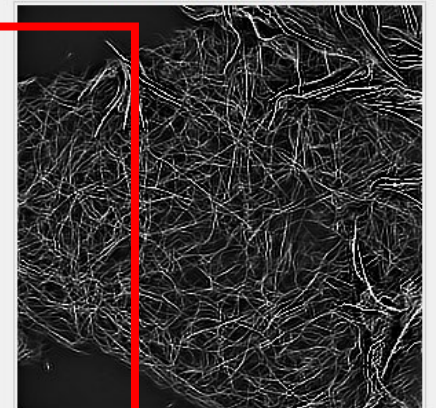
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



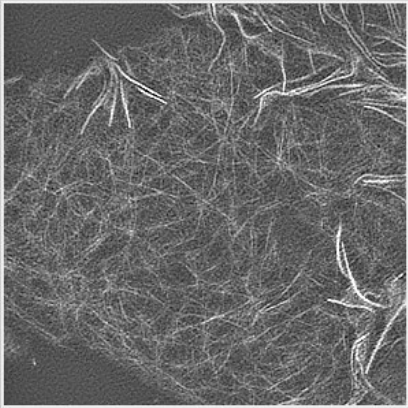
Display Range: Reset Range Reset View Export Results

▶ Frame: 1/10

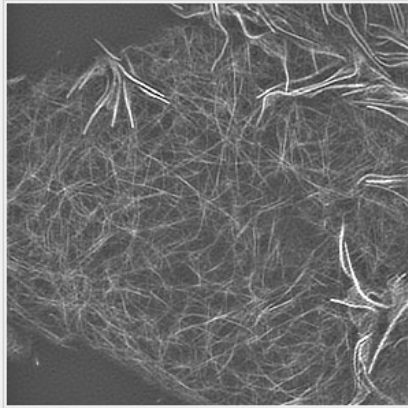
Hybrid-SIM Reconstruction Results Visualization ---- Zoom View

Reconstruction Results

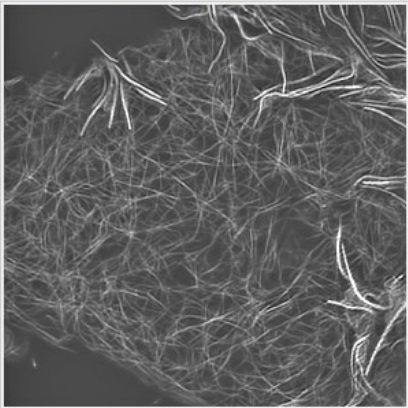
BF-SIM Reconstruction



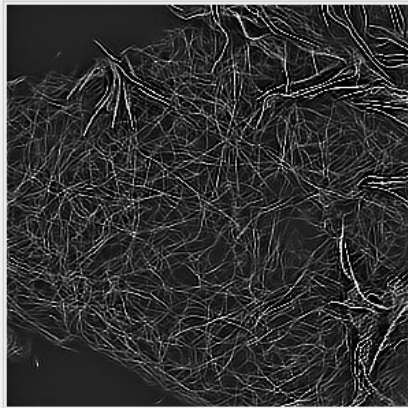
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



Display Range:

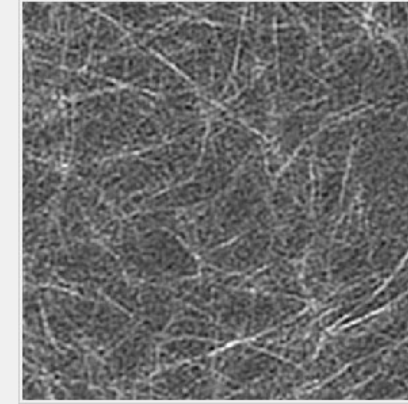
Frame: 3/10

Please scroll the mouse wheel within any area of these four images to zoom the view.

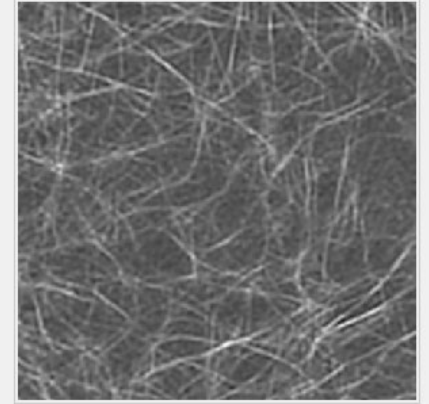
Please click the button to reset the image zoom view.

Reconstruction Results

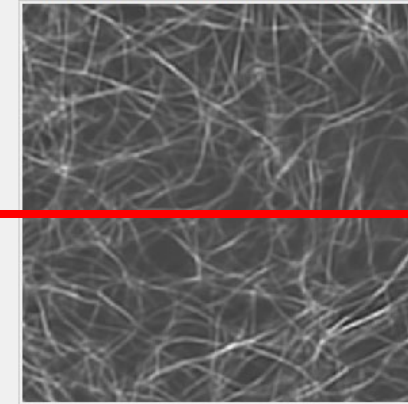
BF-SIM Reconstruction



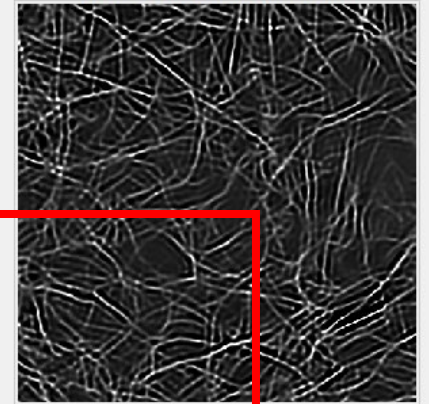
BF+Hessian-SIM Reconstruction



BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



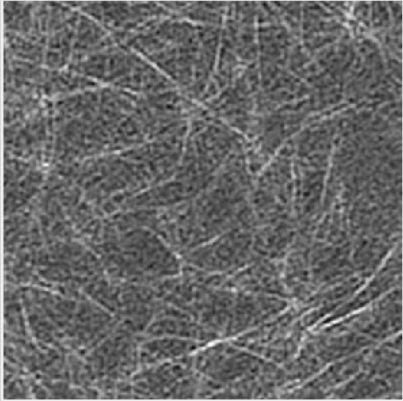
Display Range:

Frame: 3/10

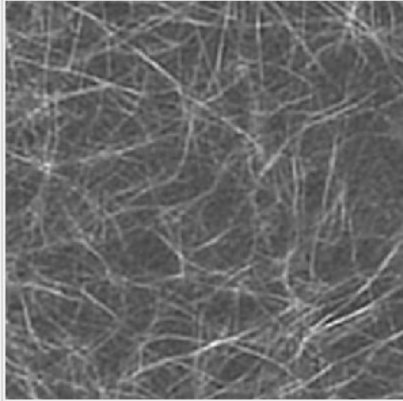
Hybrid-SIM Reconstruction Results Export

Reconstruction Results

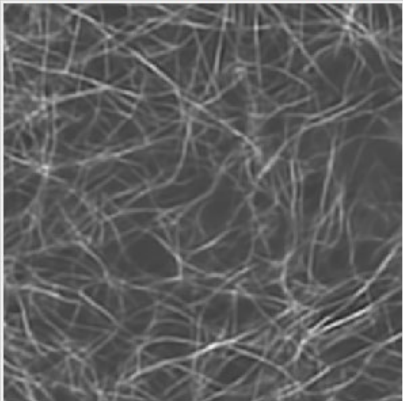
BF-SIM Reconstruction



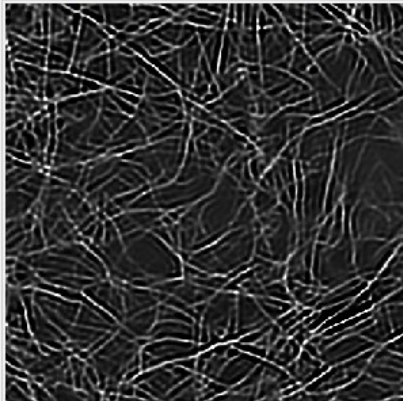
BF+Hessian-SIM Reconstruction

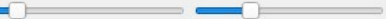


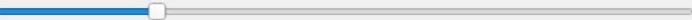
BF+Hessian+TDV-SIM Reconstruction



Hybrid-SIM Reconstruction



Display Range:  Reset Range Reset View **Export Results**

Frame:  3/10

Export Results

Export Full Reconstruction

- ☐ BF-SIM Reconstruction
- ☐ BF+Hessian-SIM Reconstruction
- ☐ BF+Hessian+TDV-SIM Reconstruction
- ☒ Hybrid-SIM Reconstruction

Export Single-Frame Comparison

- ☐ Export Comparison

Frame index: 1 Range: 1 - 10

Note: The comparison will sequentially contain BF, BF+Hessian, BF+Hessian+TDV and Hybrid reconstructions.

OK Cancel

② Please choose to export full reconstruction results of specific methods here.

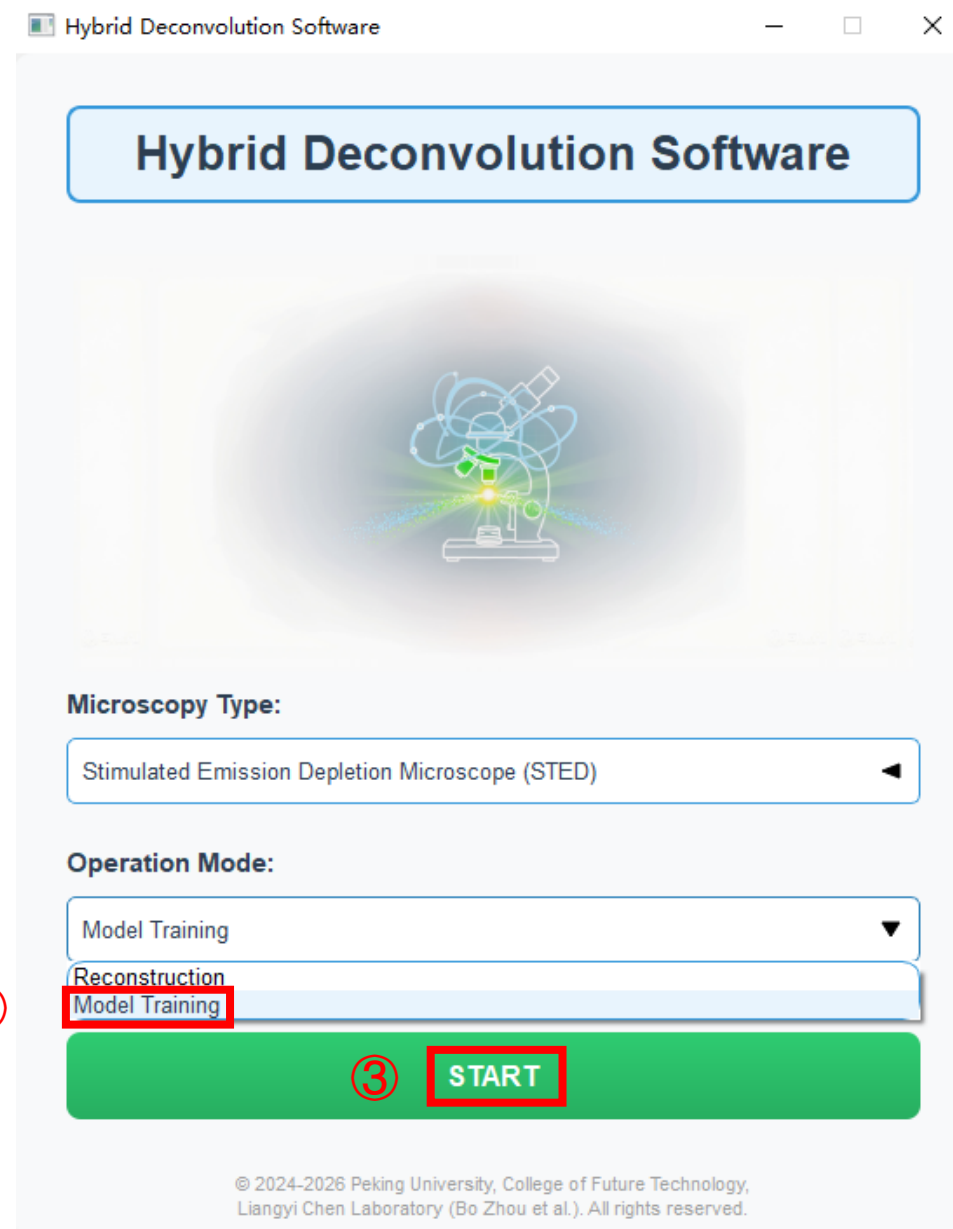
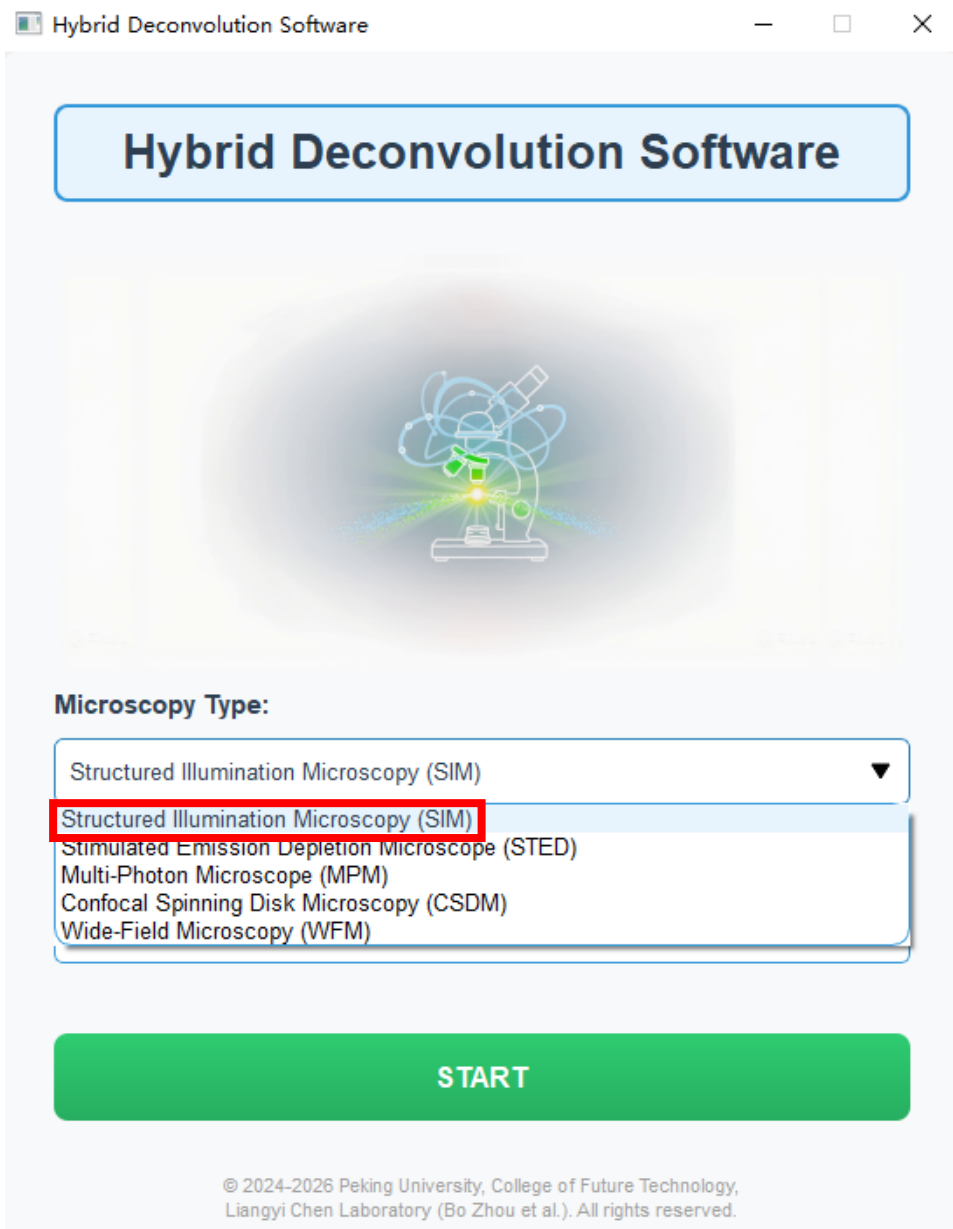
③ Please choose to export specific frame reconstruction results of all methods here.

④ OK Cancel

①

Hybrid-SIM Model Training

Start Hybrid-SIM Model Training GUI



Hybrid-SIM Training GUI layout

① Select parameters

The GUI is titled "Hybrid-SIM Training" and is divided into two main panels. The left panel, "Hybrid-SIM Training Parameters", contains three tabs: "BF-SIM", "Dataset Generation", and "TDV-DNN Training". The "BF-SIM" tab is active, showing parameters for SIM Raw Data Folder Path, SIM Excitation NA, SIM Emission NA, SIM Excitation Wavelength (nm), SIM Emission Wavelength (nm), SIM Raw Pixel Size (nm), and SIM Recon OTF Path. Each parameter has a text input field, a "Browse" button, and a "Reset" button. Below these parameters are dropdown menus for "Device" (set to "cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)") and "Operation" (set to "Dataset Generation & TDV-DNN Training"). At the bottom of this panel are buttons for "Default Parameters", "Load Parameters", and "Export Parameters". The right panel, "Hybrid-SIM Training Visualization", contains two sub-sections. The top section, "Training Curves", shows two line graphs: "Training Loss (x0.1)" vs "Batch Number" (0 to 1000) and "Validation Loss (x0.1)" vs "Epoch Number" (0.0 to 15.0). The bottom section, "Training Results", displays three side-by-side images: "TDV-DNN Input", "TDV-DNN Output", and "Ground Truth". Below these images are sliders for "Display Range" and "Frame" (set to 78/100), and buttons for "Reset Range", "Reset View", and "Check Model". At the bottom of the right panel is a "Training Log" table showing training progress.

Time	Epoch	Batch	Training-Loss	Time-Interval
[20:48:54]	Epoch: 0015/0050	Batch: 0055/0064	Training-Loss: 0.0152913416	Time-Interval: 11.97795s
[20:49:06]	Epoch: 0015/0050	Batch: 0056/0064	Training-Loss: 0.0204401612	Time-Interval: 11.97203s
[20:49:18]	Epoch: 0015/0050	Batch: 0057/0064	Training-Loss: 0.0164995622	Time-Interval: 12.05097s
[20:49:30]	Epoch: 0015/0050	Batch: 0058/0064	Training-Loss: 0.0153475739	Time-Interval: 11.95895s
[20:49:42]	Epoch: 0015/0050	Batch: 0059/0064	Training-Loss: 0.0188797563	Time-Interval: 11.86098s
[20:49:54]	Epoch: 0015/0050	Batch: 0060/0064	Training-Loss: 0.0191064160	Time-Interval: 11.96655s
[20:50:06]	Epoch: 0015/0050	Batch: 0061/0064	Training-Loss: 0.0176648907	Time-Interval: 11.90096s
[20:50:18]	Epoch: 0015/0050	Batch: 0062/0064	Training-Loss: 0.0173247159	Time-Interval: 12.00511s
[20:50:30]	Epoch: 0015/0050	Batch: 0063/0064	Training-Loss: 0.0210063998	Time-Interval: 11.86607s
[20:50:42]	Epoch: 0015/0050	Batch: 0064/0064	Training-Loss: 0.0129876565	Time-Interval: 12.01524s

② Run

③ Wait for progress

④ Visualize training process and export TDV-DNN model

Hybrid-SIM Training Parameters ---- BF-SIM Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path	/Demo_training_SIM	Browse	Reset
SIM Excitation NA	1.40	Reset	
SIM Emission NA	1.40	Reset	
SIM Excitation Wavelength (nm)	488.00	Reset	
SIM Emission Wavelength (nm)	525.00	Reset	
SIM Raw Pixel Size (nm)	65.00	Reset	
SIM Recon OTF Path	Simulation	Browse	Reset

#: Please prepare the raw data folder in advance.

SIM Raw Data Folder Path:

1. Folder path of the SIM raw data TIFF files#;
2. Please choose the folder via the Browse button;
3. Sample folder: [Demo_training_SIM].

SIM Excitation/Emission NA:

1. Excitation/Emission numerical aperture (NA) of the SIM optical system;
2. Please set the values slightly lower than the NA labeled on the objective lens.

SIM Excitation/Emission Wavelength:

1. Excitation/Emission wavelength during the SIM imaging;
2. Please set the values based on the properties of the fluorophore.

SIM Raw Pixel Size:

1. Pixel size of the SIM raw data;
2. Please set the value to the camera pixel's physical size divided by the SIM magnification ratio.

SIM Recon OTF Path:

1. TIFF file path of SIM optical transfer function path;
2. Please select an experimentally measured OTF TIFF file via the Browse button to replace the current default simulated OTF;
3. Sample experimental OTF files are in [src_Hybrid\src_Optics] subfolder within the software installation directory.

Hybrid-SIM Training Parameters ---- Dataset Generation Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

Dataset Generation Parameters

Dataset Z Average Number (frame)	20	Reset
Dataset Z Skip Number (frame)	10	Reset
Dataset XY Block Size (pixel)	256	Reset
Dataset XY Block Interval (pixel)	128	Reset
Dataset XY Poisson Noise Level	50.00	Reset
Dataset XY Gaussian Noise Level	50.00	Reset
Dataset XY Minimal Heterogeneity	0.60	Reset

Tip: The generated dataset will be saved in a new folder that appends [_Dataset] to [**SIM Raw Data Folder Path**] (e.g., [Demo_training_SIM_Dataset]).

Dataset Z Average/Skip Number:

1. Average/Skip super-resolution (SR) frame number along the temporal/axial direction of the time-lapse or volumetric data during dataset generation;
2. Please set the [Average Number] higher for generating higher SNR ground-truth (GT) images;
3. Please set the [Skip Number] higher to ensure the dataset contains a sufficient number of images.

Dataset XY Block Size/Interval:

1. Block size and interval for lateral cropping during dataset generation;
2. Please set the [Block Interval] lower to ensure the dataset contains a sufficient number of images.

Dataset XY Poisson/Gaussian Noise Level:

1. Level of Poisson/Gaussian noise added to the averaged SIM raw data during dataset generation;
2. Please set the values higher to improve model denoising capacity;
3. Please set the values lower to avoid model overfitting.

Dataset XY Minimal Heterogeneity:

1. The heterogeneity of all images in the dataset need to exceed this threshold to ensure sufficient sample information;
2. Please set the value higher for dense samples (like actin filaments) and lower for sparse samples (like peroxisomes).

Hybrid-SIM Training Parameters ---- TDV-DNN Training Parameters

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

TDV-DNN Training Parameters

Training System

100XSIM

Reset

Training Sample

Actin

Reset

Training Batch Size

10

Reset

Training Epoch Number

50

Reset

Training Learning Rate

5.00

Reset

Training System/Sample:

1. Imaging system/sample name of the raw data;
2. The trained model will be saved with a filename that includes the system name, sample name, and training timestamp. Such as the [TDV_100XSIM_Actin_20260116_092521.pth], where [100XSIM] is the system, [Actin] is the sample, [20260116_092521] represents the training start time (YYYYMMDD_HHMMSS), and [.pth] is the file format.

Training Batch Size/Epoch Number/Learning Rate:

1. Batch size, epoch number and initial learning rate of the TDV-DNN training process;
2. Please configure the [Batch Size] to maximize video random access memory usage (VRAM) without overflow;
3. Please configure the [Learning Rate] to balance training efficiency and stability.

Hybrid-SIM Training Parameters ---- Device

Please select the computing device name (cpu/cuda) for training.

Hybrid-SIM Training

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path

./Demo_training_SIM

Browse

Reset

SIM Excitation NA

1.40

Reset

SIM Emission NA

1.40

Reset

SIM Excitation Wavelength (nm)

488.00

Reset

SIM Emission Wavelength (nm)

525.00

Reset

SIM Raw Pixel Size (nm)

65.00

Reset

SIM Recon OTF Path

Simulation

Browse

Reset

Device:

cpu
cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)
cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Default Parameters

Load Parameters

Export Parameters

Start Training

Stop Training

100%

Hybrid-SIM Training Visualization

Training Curves

Training Loss ($\times 0.1$)

Batch Number

Validation Loss ($\times 0.1$)

Epoch Number

Training Results

TDV-DNN Input

TDV-DNN Output

Ground Truth

Display Range:

Reset Range

Reset View

Check Model

Frame:

78/100

Training Log

[20:48:54] Epoch: 0015/0050 | Batch: 0055/0064 | Training-Loss: 0.0152913416 | Time-Interval: 11.97795s

[20:49:06] Epoch: 0015/0050 | Batch: 0056/0064 | Training-Loss: 0.0204401612 | Time-Interval: 11.97203s

[20:49:18] Epoch: 0015/0050 | Batch: 0057/0064 | Training-Loss: 0.0164995622 | Time-Interval: 12.05097s

[20:49:30] Epoch: 0015/0050 | Batch: 0058/0064 | Training-Loss: 0.0153475739 | Time-Interval: 11.95895s

[20:49:42] Epoch: 0015/0050 | Batch: 0059/0064 | Training-Loss: 0.0188797563 | Time-Interval: 11.86098s

[20:49:54] Epoch: 0015/0050 | Batch: 0060/0064 | Training-Loss: 0.0191064160 | Time-Interval: 11.96655s

[20:50:06] Epoch: 0015/0050 | Batch: 0061/0064 | Training-Loss: 0.0176648907 | Time-Interval: 11.90096s

[20:50:18] Epoch: 0015/0050 | Batch: 0062/0064 | Training-Loss: 0.0173247159 | Time-Interval: 12.00511s

[20:50:30] Epoch: 0015/0050 | Batch: 0063/0064 | Training-Loss: 0.0210063998 | Time-Interval: 11.86607s

[20:50:42] Epoch: 0015/0050 | Batch: 0064/0064 | Training-Loss: 0.0129876565 | Time-Interval: 12.01524s

Hybrid-SIM Reconstruction Parameters ---- Default, Load and Export Parameters

Please click the button to export the current parameters.

Please click the button to reset the default parameters.

Please click the button to load the previously exported parameters.

Hybrid-SIM Training Parameters

BF-SIM

Dataset Generation

TDV-DNN Training

BF-SIM Parameters

SIM Raw Data Folder Path

./Demo_training_SIM

Browse

Reset

SIM Excitation NA

1.40

Reset

SIM Emission NA

1.40

Reset

SIM Excitation Wavelength (nm)

488.00

Reset

SIM Emission Wavelength (nm)

525.00

Reset

SIM Raw Pixel Size (nm)

65.00

Reset

SIM Recon OTF Path

Simulation

Browse

Reset

Device:

cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)

Detected: 3 device(s)

Operation:

Dataset Generation & TDV-DNN Training

Default Parameters

Load Parameters

Export Parameters

Start Training

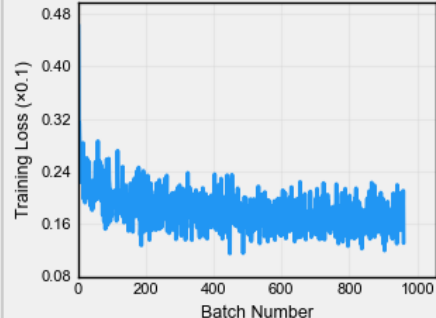
Stop Training

100%

Hybrid-SIM Training Visualization

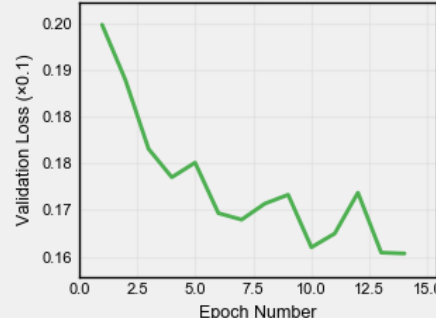
Training Curves

Training Loss ($\times 0.1$)



Batch Number

Validation Loss ($\times 0.1$)



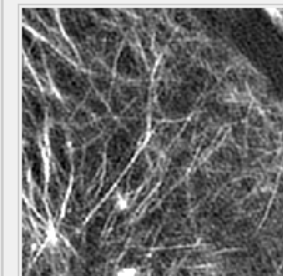
Epoch Number

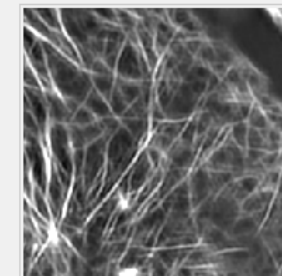
Training Results

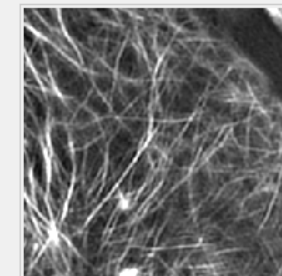
TDV-DNN Input

TDV-DNN Output

Ground Truth







Display Range:

Reset Range

Reset View

Check Model

Frame:

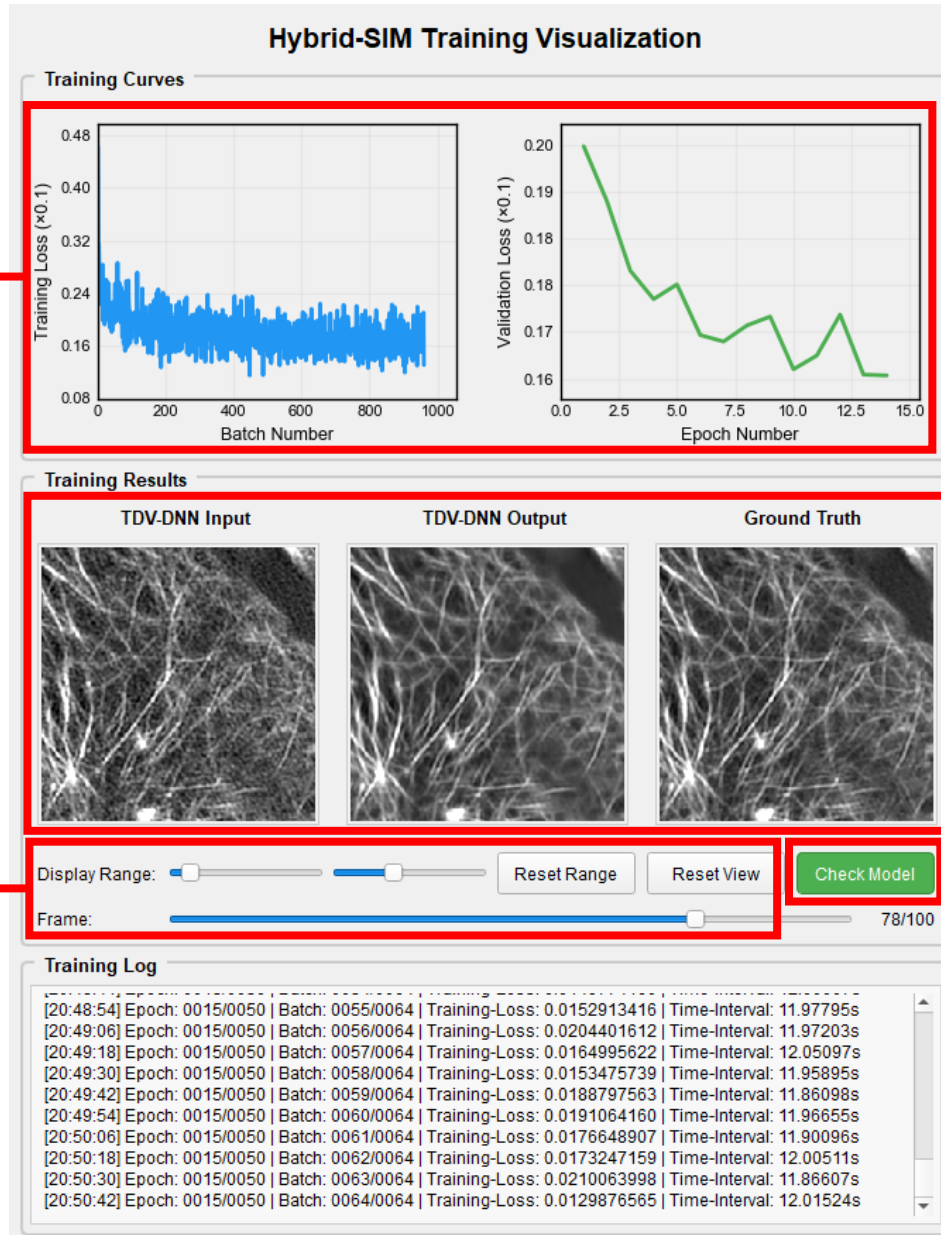
78/100

Training Log

[20:48:54]	Epoch: 0015/0050	Batch: 0055/0064	Training-Loss: 0.0152913416	Time-Interval: 11.97795s
[20:49:06]	Epoch: 0015/0050	Batch: 0056/0064	Training-Loss: 0.0204401612	Time-Interval: 11.97203s
[20:49:18]	Epoch: 0015/0050	Batch: 0057/0064	Training-Loss: 0.0164995622	Time-Interval: 12.05097s
[20:49:30]	Epoch: 0015/0050	Batch: 0058/0064	Training-Loss: 0.0153475739	Time-Interval: 11.95895s
[20:49:42]	Epoch: 0015/0050	Batch: 0059/0064	Training-Loss: 0.0188797563	Time-Interval: 11.86098s
[20:49:54]	Epoch: 0015/0050	Batch: 0060/0064	Training-Loss: 0.0191064160	Time-Interval: 11.96655s
[20:50:06]	Epoch: 0015/0050	Batch: 0061/0064	Training-Loss: 0.0176648907	Time-Interval: 11.90096s
[20:50:18]	Epoch: 0015/0050	Batch: 0062/0064	Training-Loss: 0.0173247159	Time-Interval: 12.00511s
[20:50:30]	Epoch: 0015/0050	Batch: 0063/0064	Training-Loss: 0.0210063998	Time-Interval: 11.86607s
[20:50:42]	Epoch: 0015/0050	Batch: 0064/0064	Training-Loss: 0.0129876565	Time-Interval: 12.01524s

Hybrid-SIM Training Visualization

[Training Loss] versus [Batch Number] curve (left) and, [Validation Loss] versus [Epoch Number] curve (right).



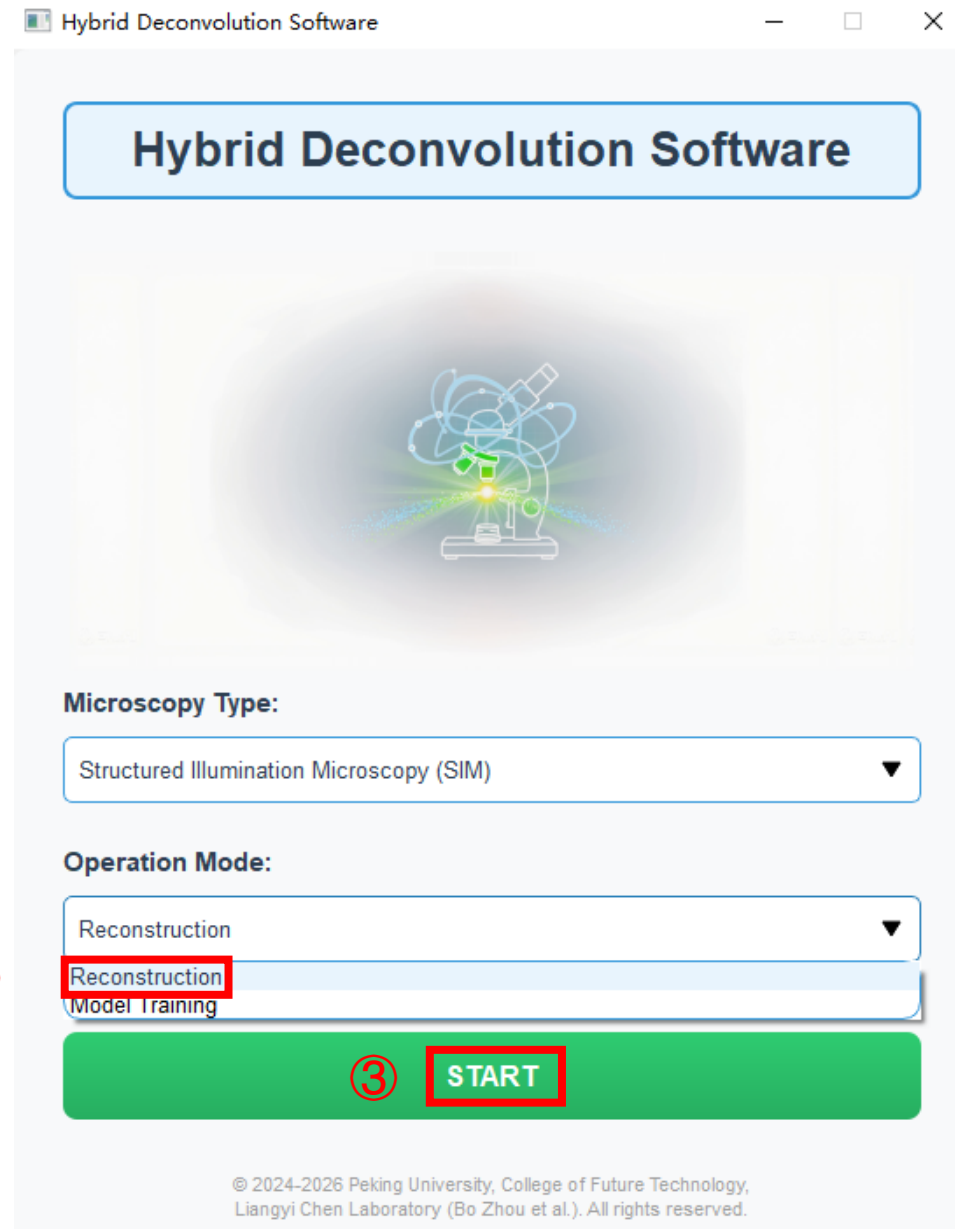
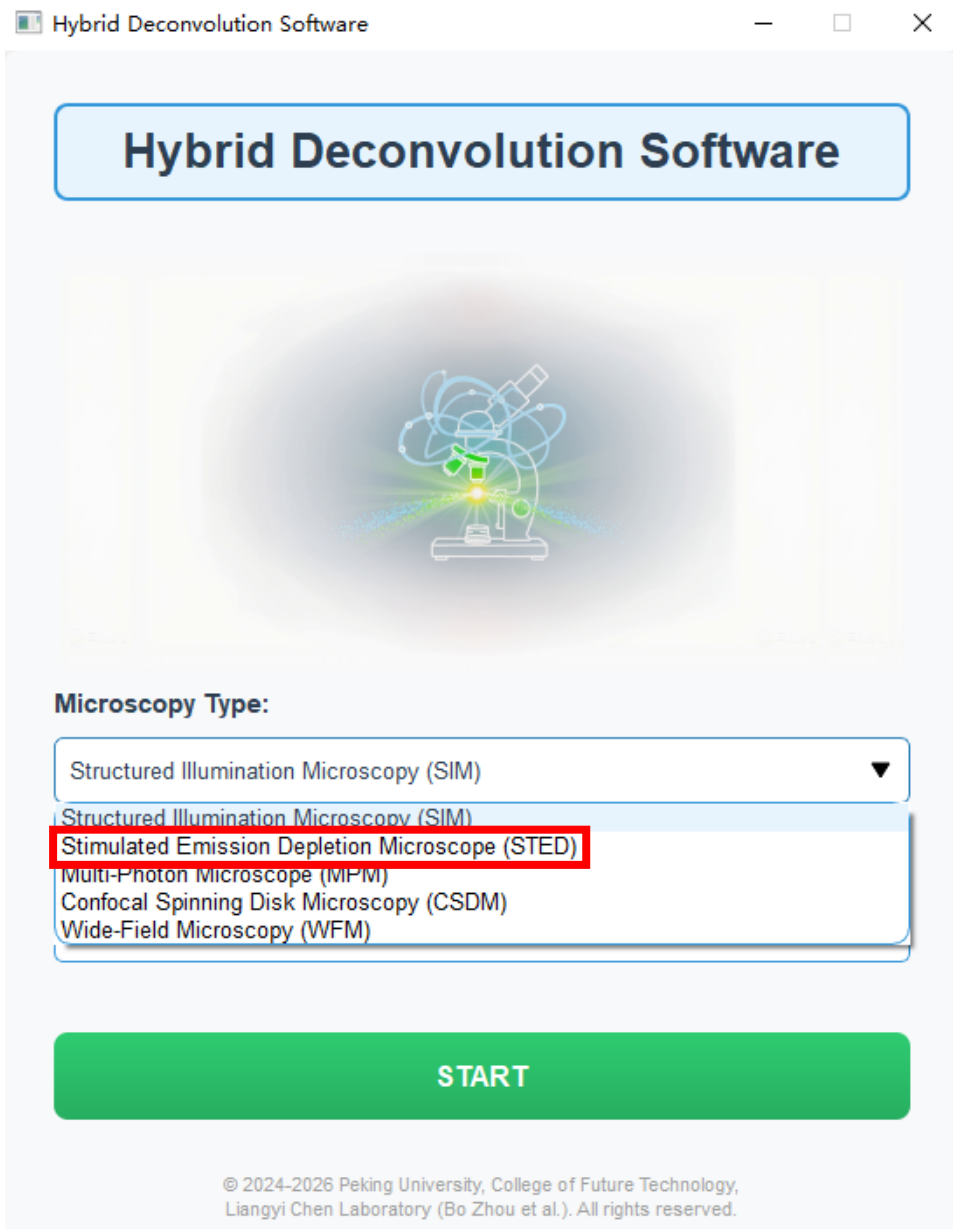
Sample input, output, and ground truth images of the TDV-DNN from the validation dataset are updated at the end of each training epoch.

Please adjust and reset the image display range, zoom view, and frame index here.

Please click the button to reveal the trained model (PTH file) in the Explorer.

Hybrid-STED Reconstruction

Start Hybrid-STED Reconstruction GUI



Hybrid-STED Reconstruction GUI Layout

① Select parameters

The screenshot shows the 'Hybrid-STED Reconstruction' GUI. The interface is divided into two main panels: 'Hybrid-STED Reconstruction Parameters' on the left and 'Reconstruction Results' on the right. The left panel has three tabs: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'Sparse Deconvolution' tab is active, showing parameters for 'Sparse Deconvolution Parameters' with input fields and 'Reset' buttons for Rolling Ball Radius (30.00), Emission Wavelength (608.00), Raw Pixel Size (15.00), Sparse NA (6.00), Sparse Offset (0.05), and Sparse Iteration Number (20). Below these is a 'Device' dropdown set to 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and buttons for 'Default Parameters', 'Load Parameters', and 'Export Parameters'. A large green 'Run Reconstruction' button is at the bottom of the left panel, with a 'Stop Reconstruction' button below it. A progress bar at the very bottom shows 100% completion with the text 'Reconstruction completed!'. The right panel, 'Reconstruction Results', displays four microscopy images: 'Hessian-STED Reconstruction', 'Hessian+TDV-STED Reconstruction', 'Hybrid-STED Background', and 'Hybrid-STED Reconstruction'. At the bottom of the right panel are sliders for 'Display Range' and 'Frame' (set to 1/10), and buttons for 'Reset Range', 'Reset View', and 'Export Results'.

Hybrid-STED Reconstruction Parameters

Hessian Denoise TDV Denoise Sparse Deconvolution

Sparse Deconvolution Parameters

Rolling Ball Radius (pixels) 30.00 Reset

Emission Wavelength (nm) 608.00 Reset

Raw Pixel Size (nm) 15.00 Reset

Sparse NA 6.00 Reset

Sparse Offset 0.05 Reset

Sparse Iteration Number 20 Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters Load Parameters Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed!

Reconstruction Results

Hessian-STED Reconstruction Hessian+TDV-STED Reconstruction

Hybrid-STED Background Hybrid-STED Reconstruction

Display Range: Reset Range Reset View Export Results

Frame: 1/10

④ Visualize and export results

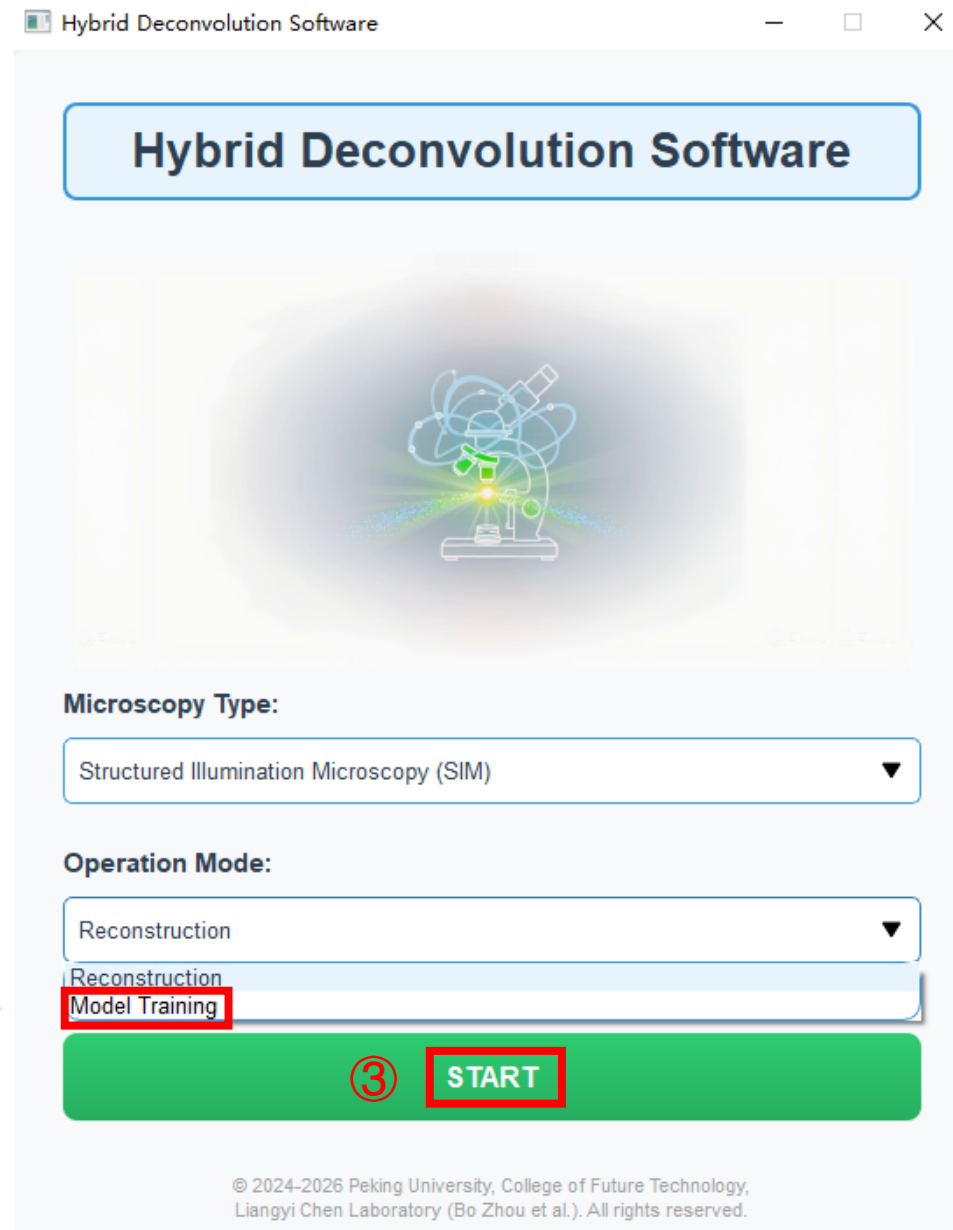
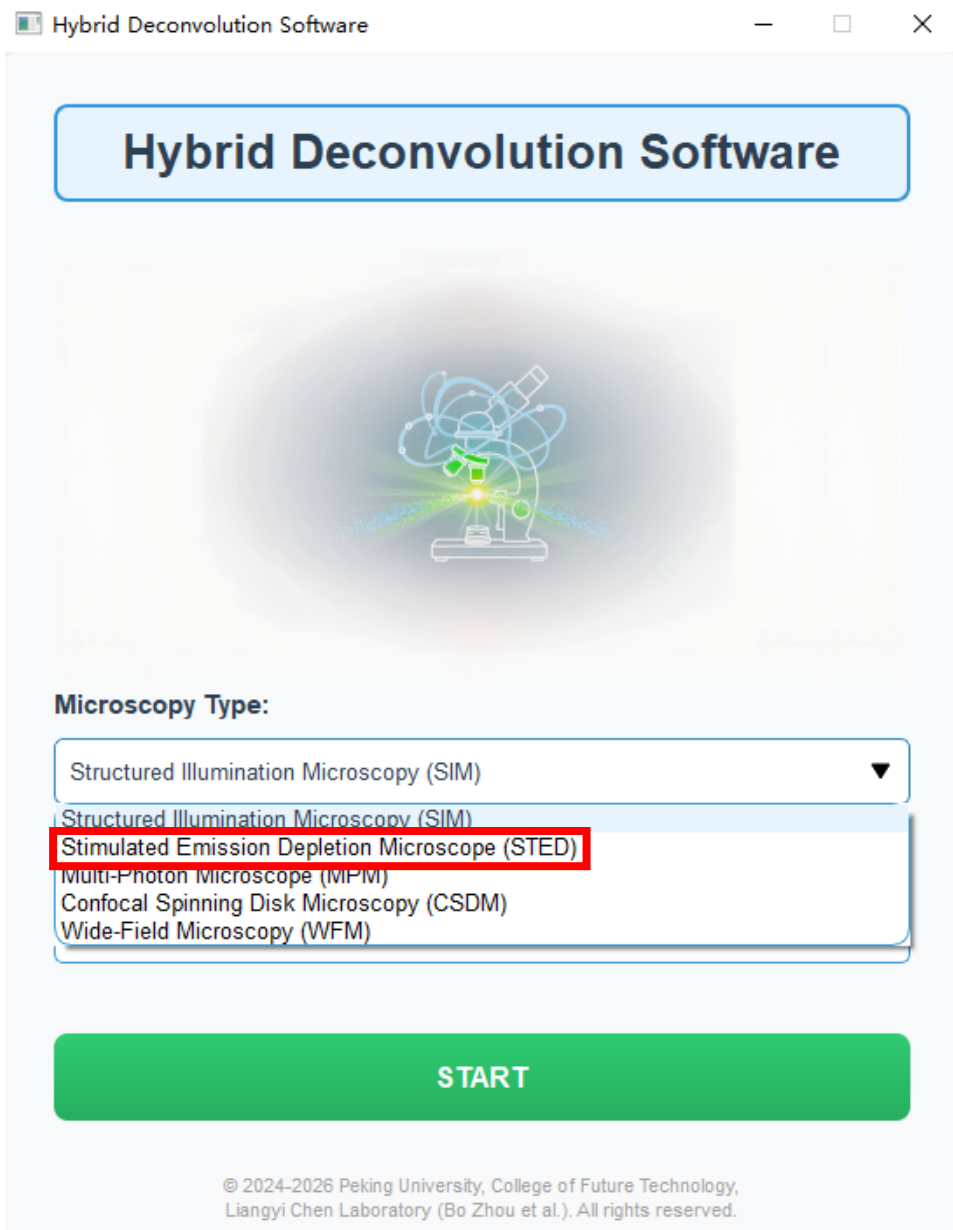
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-STED Model Training

Start Hybrid-STED Model Training GUI



Hybrid-STED Training GUI layout

① Select parameters

The screenshot displays the Hybrid-STED Training GUI, which is divided into two main panels: 'Hybrid-STED Training Parameters' on the left and 'Hybrid-STED Training Visualization' on the right.

Hybrid-STED Training Parameters Panel:

- Dataset Generation Parameters:** Includes fields for Raw Data Folder Path (set to /Demo_training_STED), Emission NA (5.00), Emission Wavelength (nm) (608.00), Raw Pixel Size (nm) (15.00), Dataset Z Average Number (frame) (1), Dataset Z Skip Number (frame) (1), Dataset XY Block Size (pixel) (256), Dataset XY Block Interval (pixel) (64), Dataset XY Poisson Noise Level (300.00), Dataset XY Gaussian Noise Level (300.00), and Dataset XY Minimal Heterogeneity (0.60). Each field has a 'Reset' button.
- Device:** A dropdown menu showing 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' with a 'Detected: 3 device(s)' label.
- Operation:** A dropdown menu showing 'Dataset Generation & TDV-DNN Training'.
- Buttons:** 'Default Parameters', 'Load Parameters', 'Export Parameters', 'Start Training' (highlighted with a red box), and 'Stop Training'.
- Progress Bar:** A progress bar at the bottom shows '4%' completion.

Hybrid-STED Training Visualization Panel:

- Training Curves:** Two line graphs. The left graph shows 'Training Loss (x0.1)' vs 'Batch Number' (0 to 125), with a blue line showing a sharp drop and then leveling off. The right graph shows 'Validation Loss (x0.1)' vs 'Epoch Number' (0.0 to 2.0), with a green line showing a steady decrease.
- Training Results:** Three side-by-side images labeled 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth', showing a microscopic image of a cell.
- Display Range:** A slider bar with 'Reset Range' and 'Reset View' buttons.
- Frame:** A slider bar with a 'Check Model' button.
- Training Log:** A text area displaying a log of training progress, including epoch numbers, batch numbers, training loss, and time intervals.

② Run

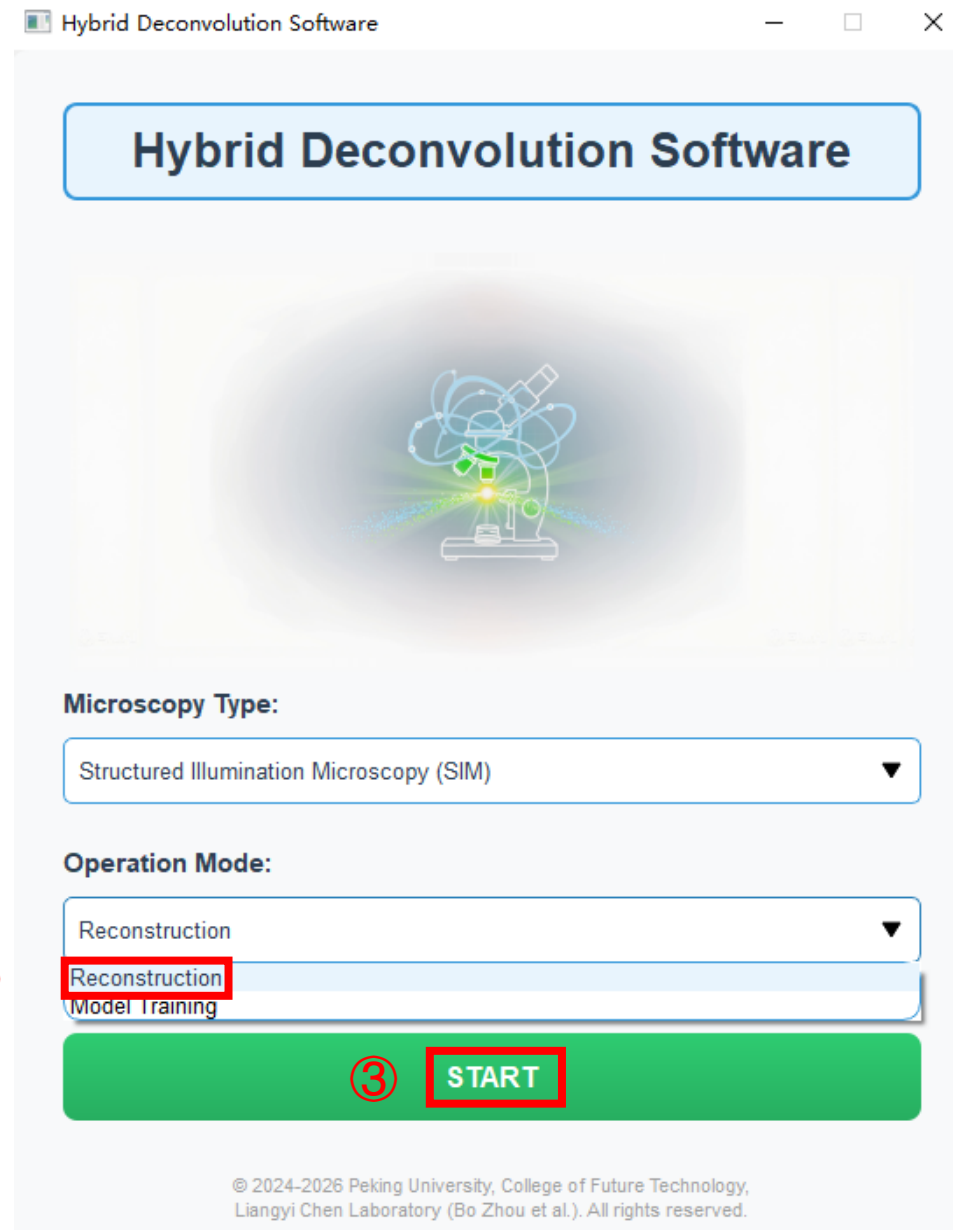
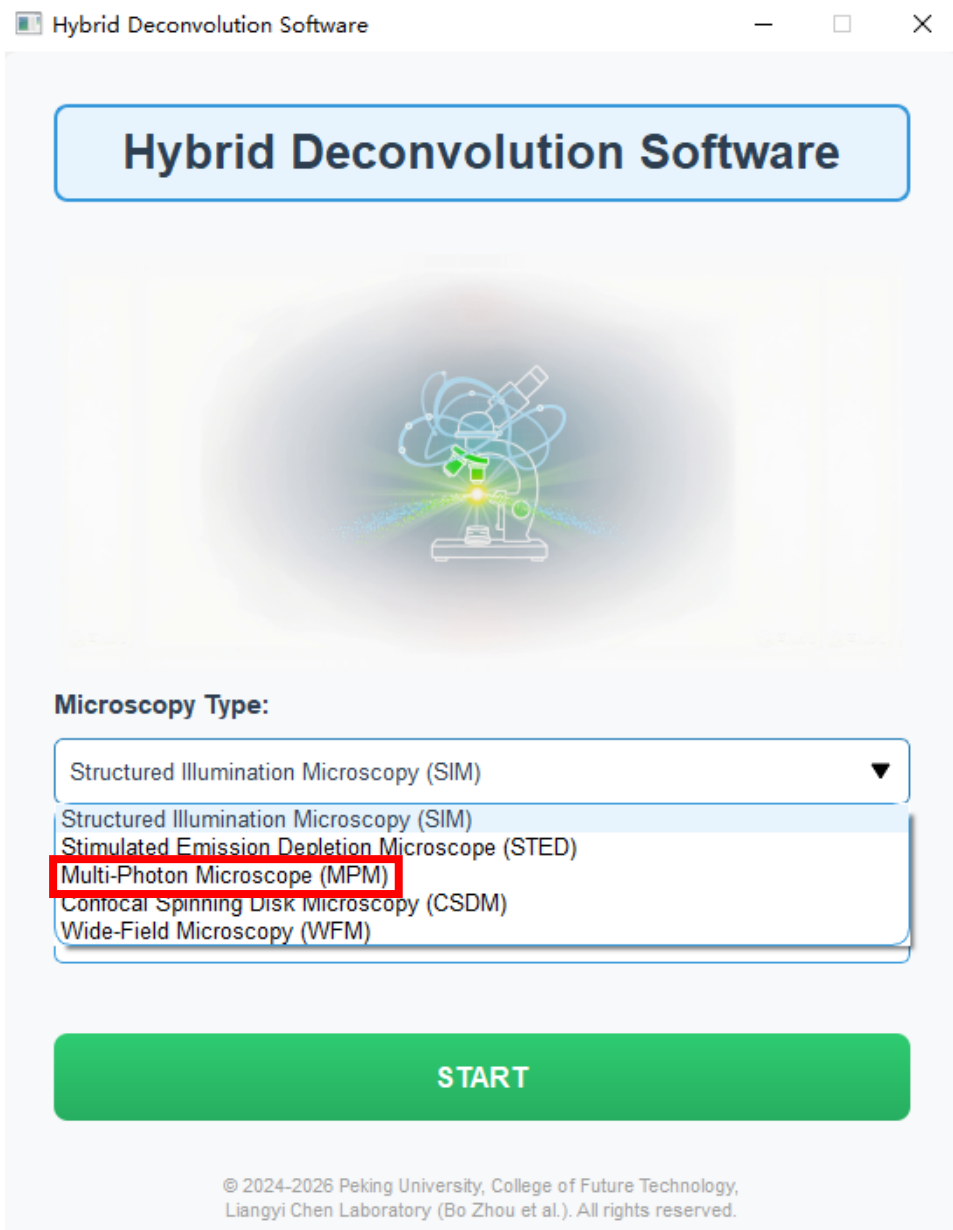
③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-MPM Reconstruction

Start Hybrid-MPM Reconstruction GUI



Hybrid-MPM Reconstruction GUI Layout

① Select parameters

The screenshot displays the 'Hybrid-MPM Reconstruction' application window. The interface is divided into two main sections: 'Hybrid-MPM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right.

Hybrid-MPM Reconstruction Parameters:

- Three tabs are visible: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution' (which is selected and highlighted with a green underline).
- Under the 'Sparse Deconvolution Parameters' section, there are six input fields, each with a 'Reset' button:
 - Rolling Ball Radius (pixels): 10.00
 - Emission Wavelength (nm): 525.00
 - Raw Pixel Size (nm): 200.00
 - Sparse NA: 0.30
 - Sparse Offset: 0.02
 - Sparse Iteration Number: 25
- A red box highlights the 'Rolling Ball Radius' field, with a text box explaining: **Rolling Ball Radius:** Radius of curvature of the paraboloid during rolling ball background filtering.
- At the bottom, a 'Device' dropdown menu shows 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and 'Detected: 3 device(s)'. Below this are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'.
- A large green button labeled 'Run Reconstruction' is prominently displayed, with a red box around it and an arrow pointing to it from the 'Run' instruction.
- Below the 'Run' button is a grey button labeled 'Stop Reconstruction'.
- At the bottom, a green progress bar shows '100%' completion, with the text 'Reconstruction completed!' below it. An arrow points to this section from the 'Wait for progress' instruction.

Reconstruction Results:

- This section contains four image panels:
 - 'Hessian-MPM Reconstruction': Shows a noisy, low-contrast image of a cellular structure.
 - 'Hessian+TDV-MPM Reconstruction': Shows a clearer, higher-contrast image of the same structure.
 - 'Hybrid-MPM Background': Shows a blurred, low-contrast background image.
 - 'Hybrid-MPM Reconstruction': Shows the final reconstructed image, which is sharp and clear.
- Below the images are controls for 'Display Range' (a slider), 'Reset Range', 'Reset View', and 'Export Results' (a blue button).
- At the bottom, there is a 'Frame' slider showing '62/100'.

④ Visualize and export results

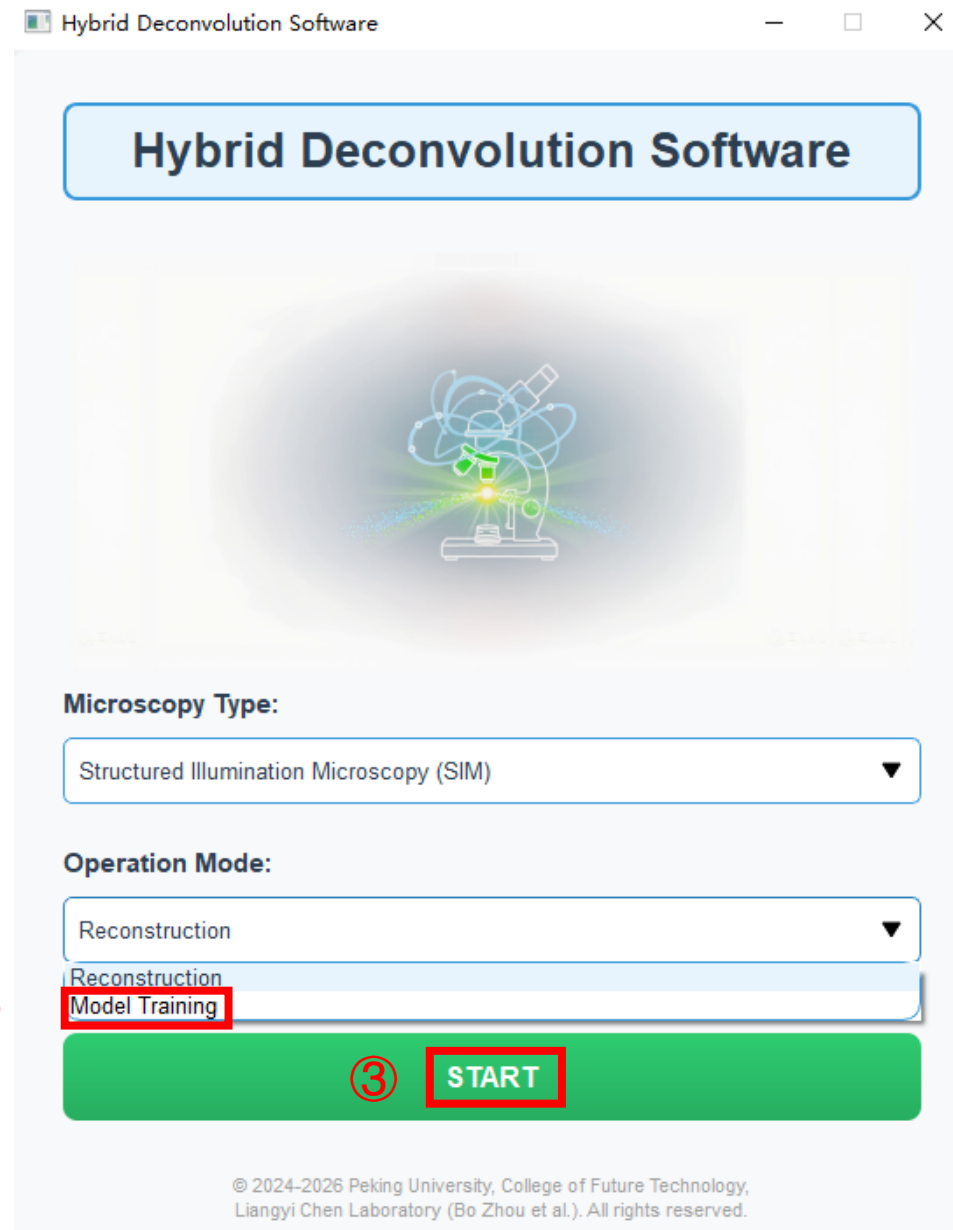
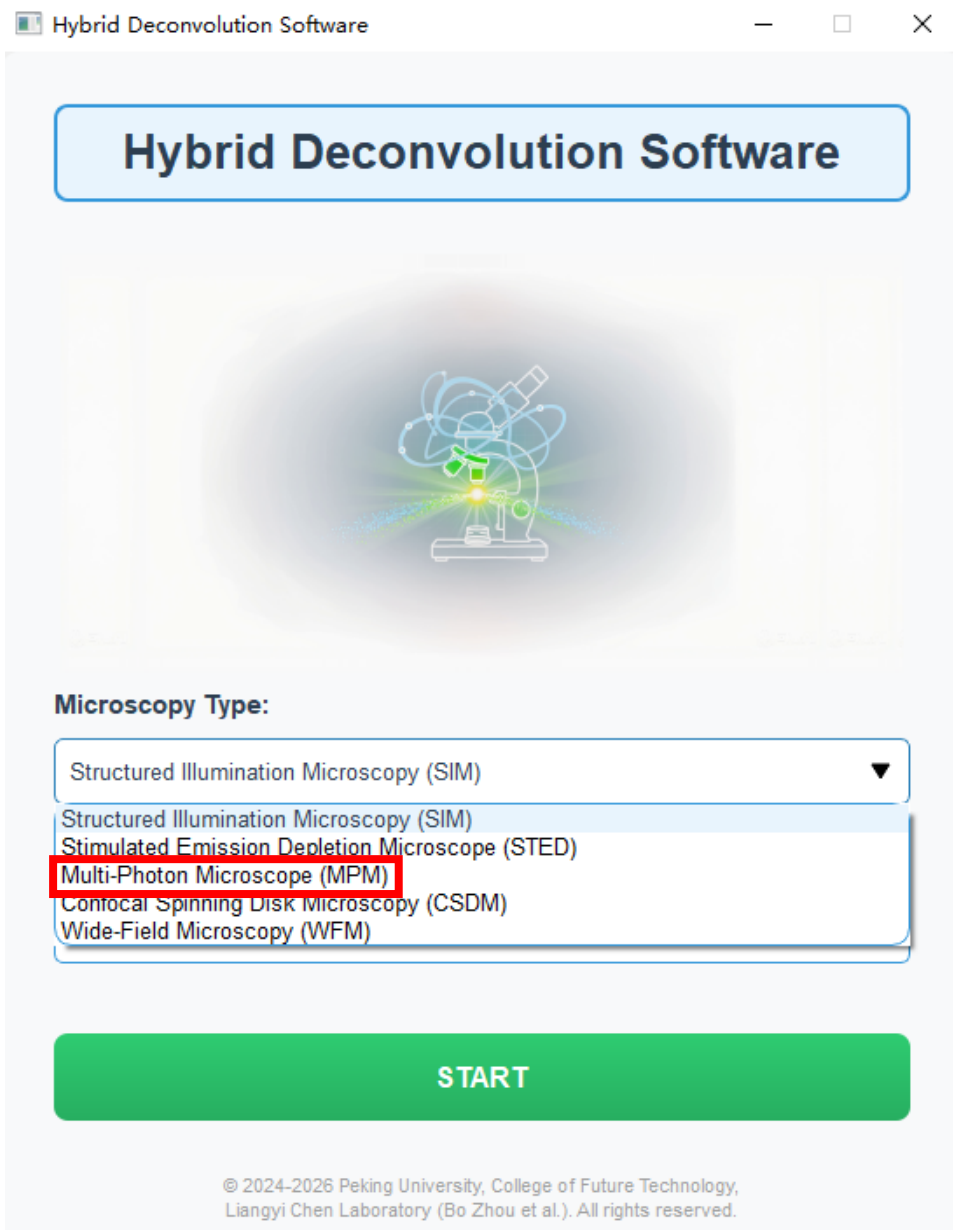
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-MPM Model Training

Start Hybrid-MPM Model Training GUI



Hybrid-MPM Training GUI layout

① Select parameters

The screenshot displays the Hybrid-MPM Training GUI, which is divided into two main panels: 'Hybrid-MPM Training Parameters' on the left and 'Hybrid-MPM Training Visualization' on the right.

Hybrid-MPM Training Parameters Panel:

- Dataset Generation Parameters:** Includes fields for Raw Data Folder Path (./Demo_training_MPM), Emission NA (0.60), Emission Wavelength (nm) (525.00), Raw Pixel Size (nm) (200.00), Dataset Z Average Number (frame) (20), Dataset Z Skip Number (frame) (10), Dataset XY Block Size (pixel) (256), Dataset XY Block Interval (pixel) (64), Dataset XY Poisson Noise Level (300.00), Dataset XY Gaussian Noise Level (300.00), and Dataset XY Minimal Heterogeneity (0.60). Each field has a 'Reset' button.
- Device and Operation:** A dropdown menu for 'Device' (cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)) and a dropdown for 'Operation' (TDV-DNN Training).
- Buttons:** 'Default Parameters', 'Load Parameters', 'Export Parameters', 'Start Training' (highlighted with a red box), and 'Stop Training'.
- Progress Bar:** A progress bar at the bottom shows 10% completion.

Hybrid-MPM Training Visualization Panel:

- Training Curves:** Two line graphs. The left graph shows 'Training Loss (x0.1)' vs 'Batch Number' (0 to 400), with a blue line showing a sharp drop from 1.25 to near 0. The right graph shows 'Validation Loss (x0.1)' vs 'Epoch Number' (0 to 5), with a green line showing a general downward trend from 0.04 to 0.03.
- Training Results:** Three side-by-side images labeled 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth', showing microscopic cell images.
- Display Range and Frame:** A 'Display Range' slider and a 'Frame' slider (set to 36/100) are present, along with 'Reset Range', 'Reset View', and 'Check Model' buttons.
- Training Log:** A scrollable text area showing the training progress, including epoch and batch numbers, training loss, and time intervals.

② Run

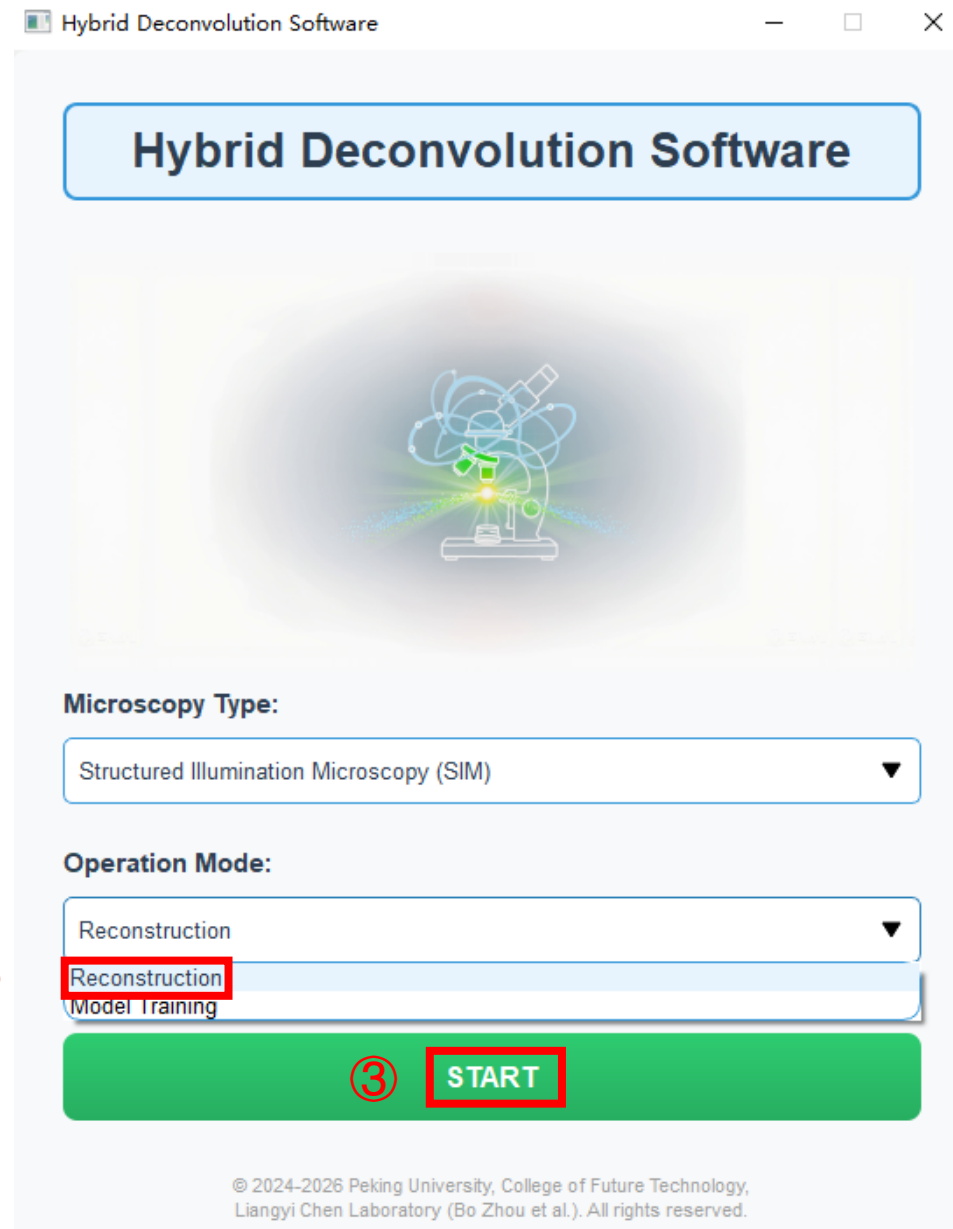
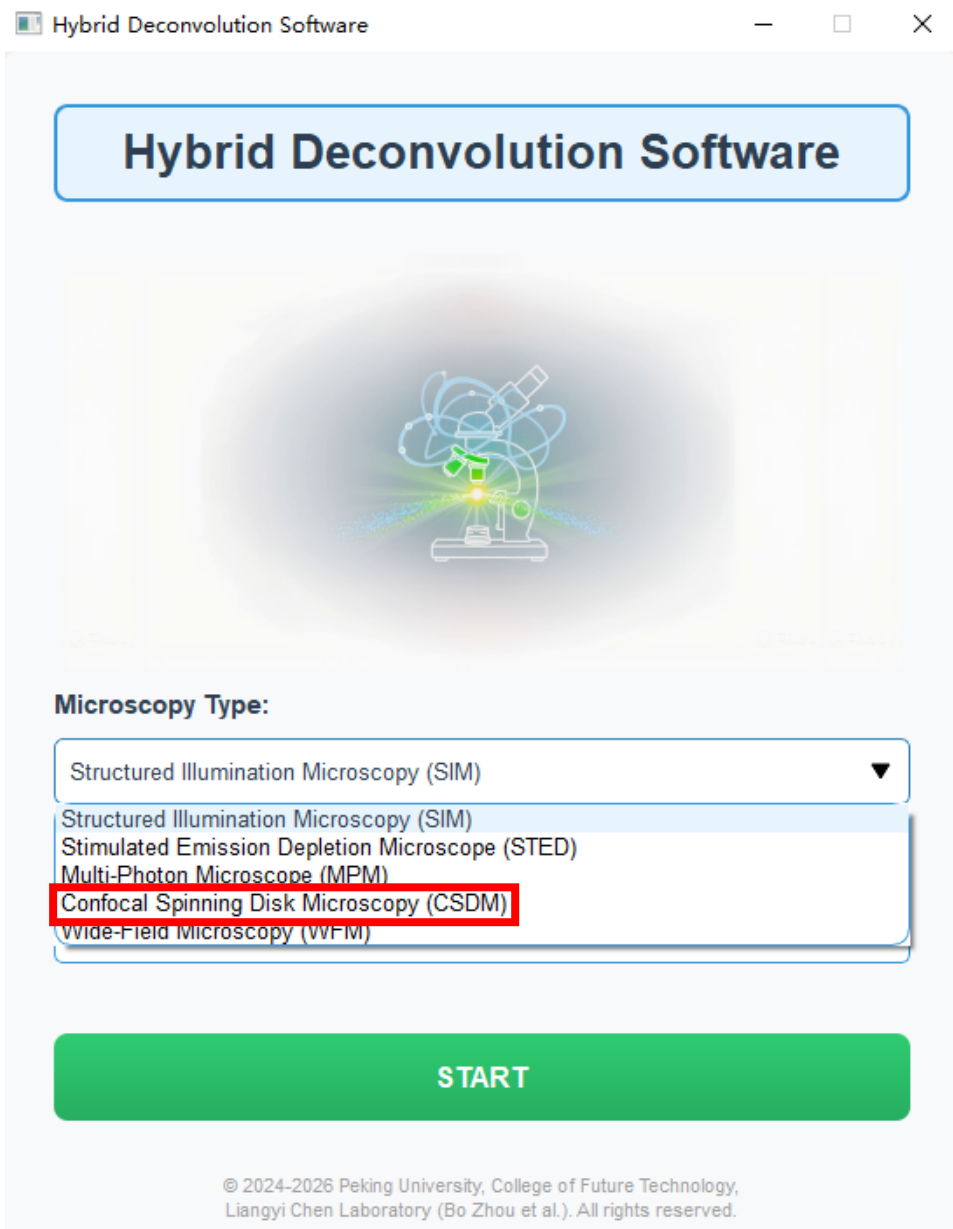
③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-CSDM Reconstruction

Start Hybrid-CSDM Reconstruction GUI



Hybrid-CSDM Reconstruction GUI Layout

① Select parameters

The screenshot shows the 'Hybrid-CSDM Reconstruction' GUI. The interface is divided into two main panels: 'Hybrid-CSDM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right. The left panel has three tabs: 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution' (which is selected). Under the 'Sparse Deconvolution' tab, there are several input fields with numerical values and 'Reset' buttons: 'Rolling Ball Radius (pixels)' (100.00), 'Emission Wavelength (nm)' (525.00), 'Raw Pixel Size (nm)' (65.00), 'Sparse NA' (1.20), 'Sparse Offset' (0.05), and 'Sparse Iteration Number' (20). Below these fields, there is a 'Device' dropdown menu showing 'cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)' and a 'Detected: 3 device(s)' label. At the bottom of the left panel are three buttons: 'Default Parameters', 'Load Parameters', and 'Export Parameters'. The right panel, titled 'Reconstruction Results', displays four grayscale images in a 2x2 grid. The top-left image is 'Hessian-CSDM Reconstruction', the top-right is 'Hessian+TDV-CSDM Reconstruction', the bottom-left is 'Hybrid-CSDM Background', and the bottom-right is 'Hybrid-CSDM Reconstruction'. Below the images are two sliders for 'Display Range' and 'Frame' (set to 38/100), along with 'Reset Range', 'Reset View', and 'Export Results' buttons. A green progress bar at the bottom of the left panel shows 100% completion, with the text 'Reconstruction completed!' below it. A red box highlights the 'Run Reconstruction' button in the left panel. Red arrows point from the numbered instructions to their corresponding GUI elements.

Hybrid-CSDM Reconstruction Parameters

Hessian Denoise | TDV Denoise | **Sparse Deconvolution**

Sparse Deconvolution Parameters

Rolling Ball Radius (pixels): 100.00 [Reset]

Emission Wavelength (nm): 525.00 [Reset]

Raw Pixel Size (nm): 65.00 [Reset]

Sparse NA: 1.20 [Reset]

Sparse Offset: 0.05 [Reset]

Sparse Iteration Number: 20 [Reset]

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters | Load Parameters | Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed !

Reconstruction Results

Hessian-CSDM Reconstruction | Hessian+TDV-CSDM Reconstruction

Hybrid-CSDM Background | Hybrid-CSDM Reconstruction

Display Range: [Slider] [Reset Range] [Reset View] [Export Results]

Frame: [Slider] 38/100

④ Visualize and export results

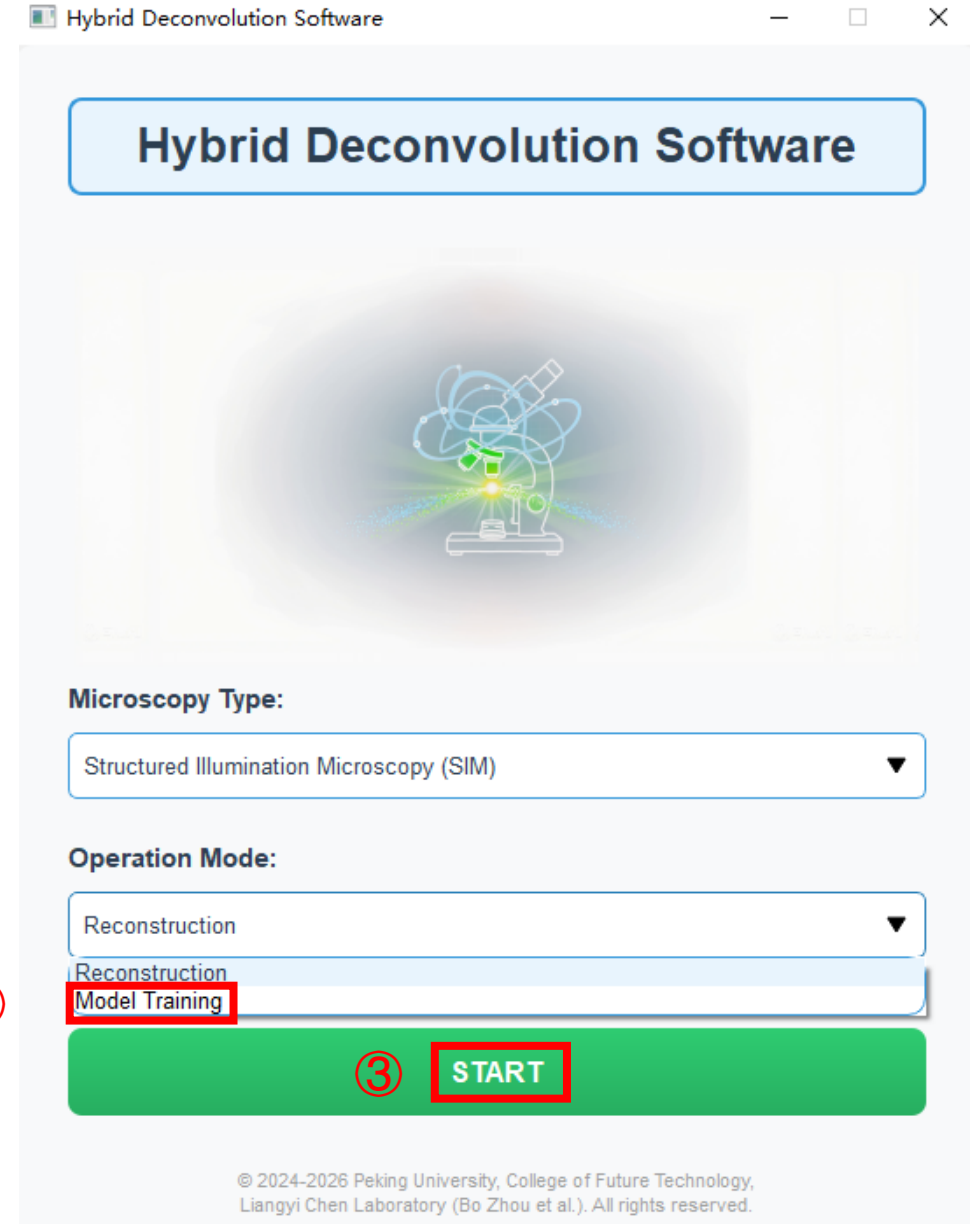
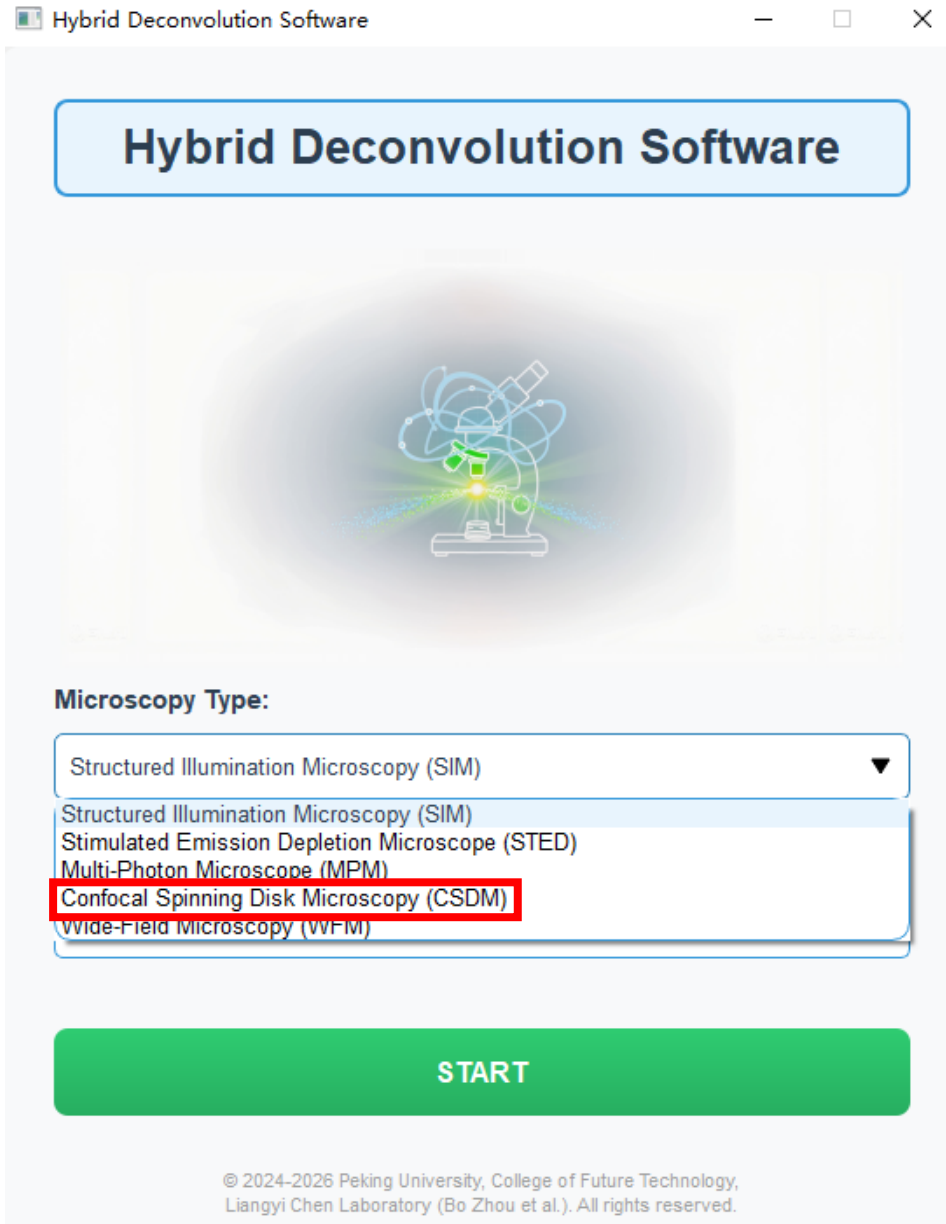
② Run

③ Wait for progress

Tip: The remaining parameters have been covered in the Hybrid-SIM section.

Hybrid-CSDM Model Training

Start Hybrid-CSDM Model Training GUI



Hybrid-CSDM Training GUI layout

① Select parameters

The screenshot displays the Hybrid-CSDM Training GUI, which is divided into two main panels: 'Hybrid-CSDM Training Parameters' on the left and 'Hybrid-CSDM Training Visualization' on the right.

Hybrid-CSDM Training Parameters Panel:

- Dataset Generation Parameters:** Includes fields for Raw Data Folder Path (Demo_training_CSDM), Emission NA (1.30), Emission Wavelength (nm) (525.00), Raw Pixel Size (nm) (65.00), Dataset Z Average Number (frame) (20), Dataset Z Skip Number (frame) (16), Dataset XY Block Size (pixel) (256), Dataset XY Block Interval (pixel) (128), Dataset XY Poisson Noise Level (200.00), Dataset XY Gaussian Noise Level (200.00), and Dataset XY Minimal Heterogeneity (0.60). Each field has a 'Reset' button.
- Device and Operation:** A dropdown menu for 'Device' (cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB)) and a dropdown for 'Operation' (Dataset Generation & TDV-DNN Training).
- Buttons:** 'Default Parameters', 'Load Parameters', 'Export Parameters', 'Start Training' (highlighted with a red box), and 'Stop Training'.
- Progress Bar:** A progress bar at the bottom shows 4% completion.

Hybrid-CSDM Training Visualization Panel:

- Training Curves:** Two line graphs. The left graph shows 'Training Loss (x0.1)' vs 'Batch Number' (0 to 125), with a blue line showing a sharp drop and then leveling off. The right graph shows 'Validation Loss (x0.1)' vs 'Epoch Number' (0.0 to 2.0), with a green line showing a steady decrease.
- Training Results:** Three side-by-side images labeled 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth', showing grayscale microscopy images of cells.
- Display Range and Frame:** A 'Display Range' slider and a 'Frame' slider (set to 52/100) are located below the images.
- Buttons:** 'Reset Range', 'Reset View', and 'Check Model'.
- Training Log:** A scrollable text area at the bottom showing training progress, including epoch, batch, training loss, validation loss, and time intervals. The log ends with 'TDV-DNN training stopped!' and 'Ready!'.

② Run

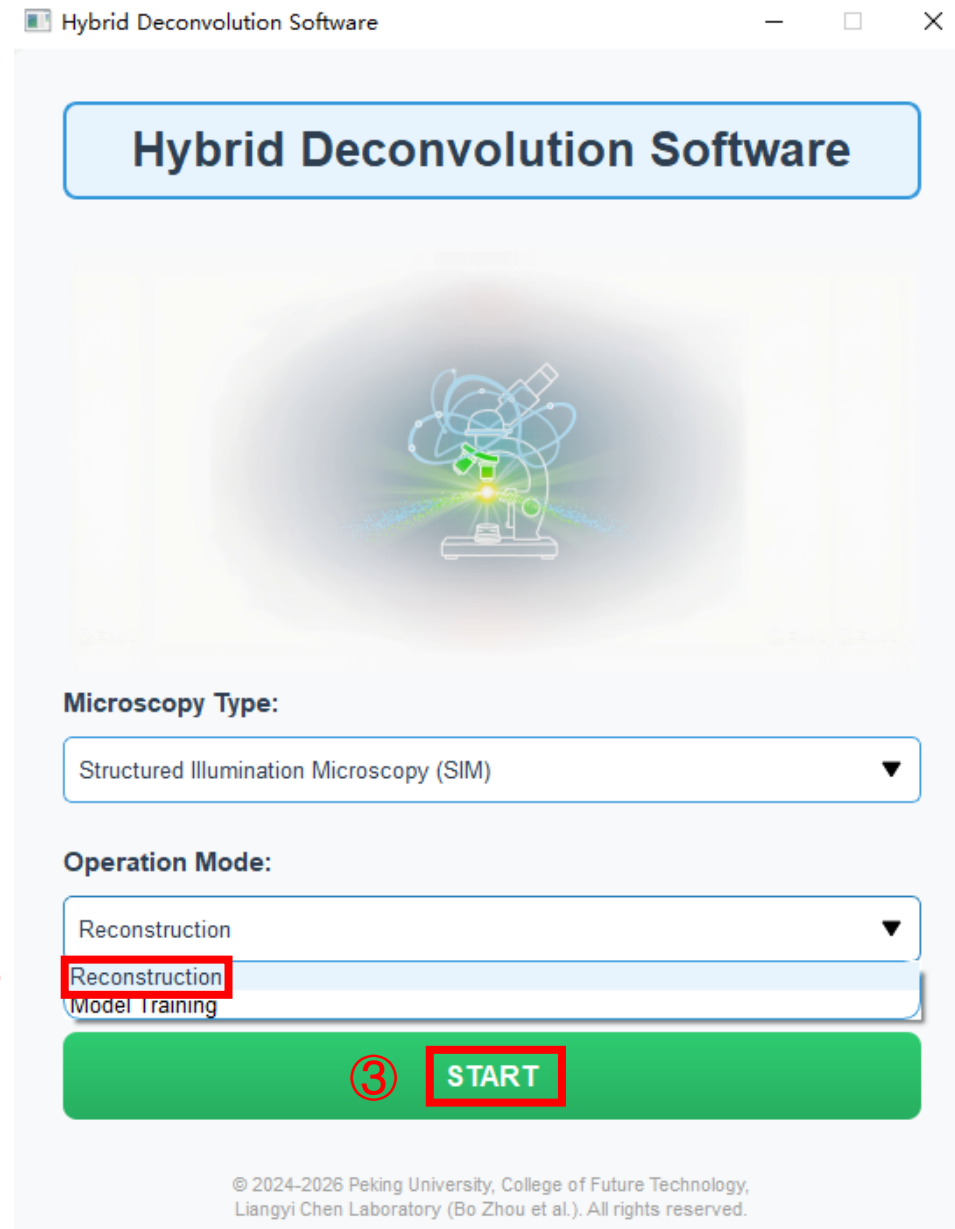
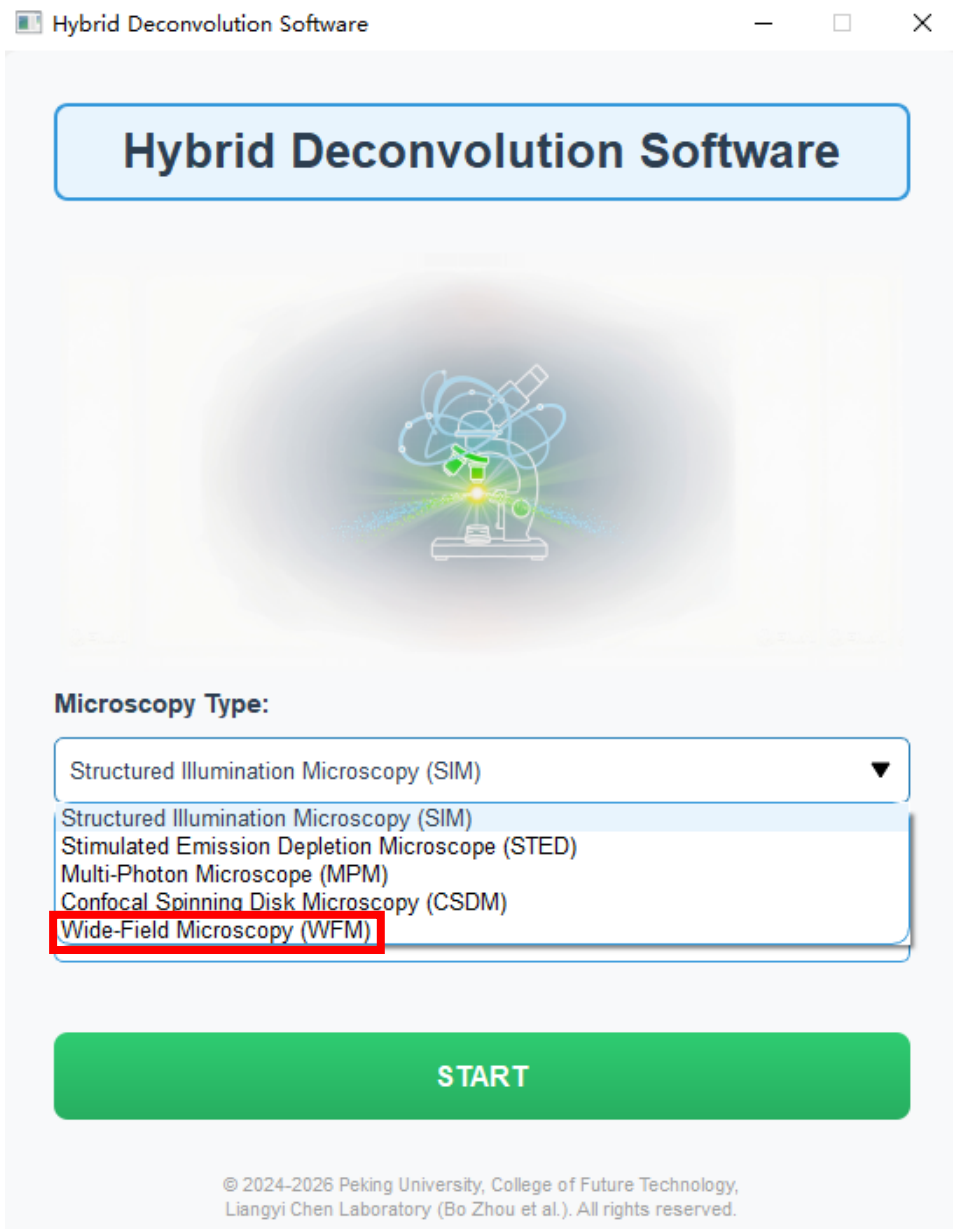
③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-WFM Reconstruction

Start Hybrid-WFM Reconstruction GUI



Hybrid-WFM Reconstruction GUI Layout

① Select parameters

The screenshot shows the 'Hybrid-WFM Reconstruction' application window. It is divided into two main sections: 'Hybrid-WFM Reconstruction Parameters' on the left and 'Reconstruction Results' on the right. The left section has tabs for 'BF-WFM', 'Hessian Denoise', 'TDV Denoise', and 'Sparse Deconvolution'. The 'BF-WFM' tab is active, showing parameters for 'BF-WFM Parameters' such as 'WFM Raw Data Path', 'WFM Raw Pixel Size (nm)', 'WFM Emission Wavelength (nm)', and 'WFM Emission NA'. Below these are buttons for 'Default Parameters', 'Load Parameters', and 'Export Parameters'. A 'Run Reconstruction' button is highlighted with a red box. Below it is a 'Stop Reconstruction' button. At the bottom, a green progress bar shows 100% completion, and the text 'Reconstruction completed!' is displayed. The right section, 'Reconstruction Results', displays four grayscale images: 'BF-WFM Reconstruction', 'BF+Hessian-WFM Reconstruction', 'BF+Hessian+TDV-WFM Reconstruction', and 'Hybrid-WFM Reconstruction'. Below the images are sliders for 'Display Range' and 'Frame' (set to 38/100), and buttons for 'Reset Range', 'Reset View', and 'Export Results'.

Hybrid-WFM Reconstruction Parameters

BF-WFM Hessian Denoise TDV Denoise Sparse Deconvolution

BF-WFM Parameters

WFM Raw Data Path o_Recon_WFM.tif Browse Reset

WFM Raw Pixel Size (nm) 65.00 Reset

WFM Emission Wavelength (nm) 525.00 Reset

WFM Emission NA 1.40 Reset

Device: cuda:0 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Default Parameters Load Parameters Export Parameters

Run Reconstruction

Stop Reconstruction

100%

Reconstruction completed!

Reconstruction Results

BF-WFM Reconstruction BF+Hessian-WFM Reconstruction

BF+Hessian+TDV-WFM Reconstruction Hybrid-WFM Reconstruction

Display Range: Reset Range Reset View Export Results

Frame: 38/100

④ Visualize and export results

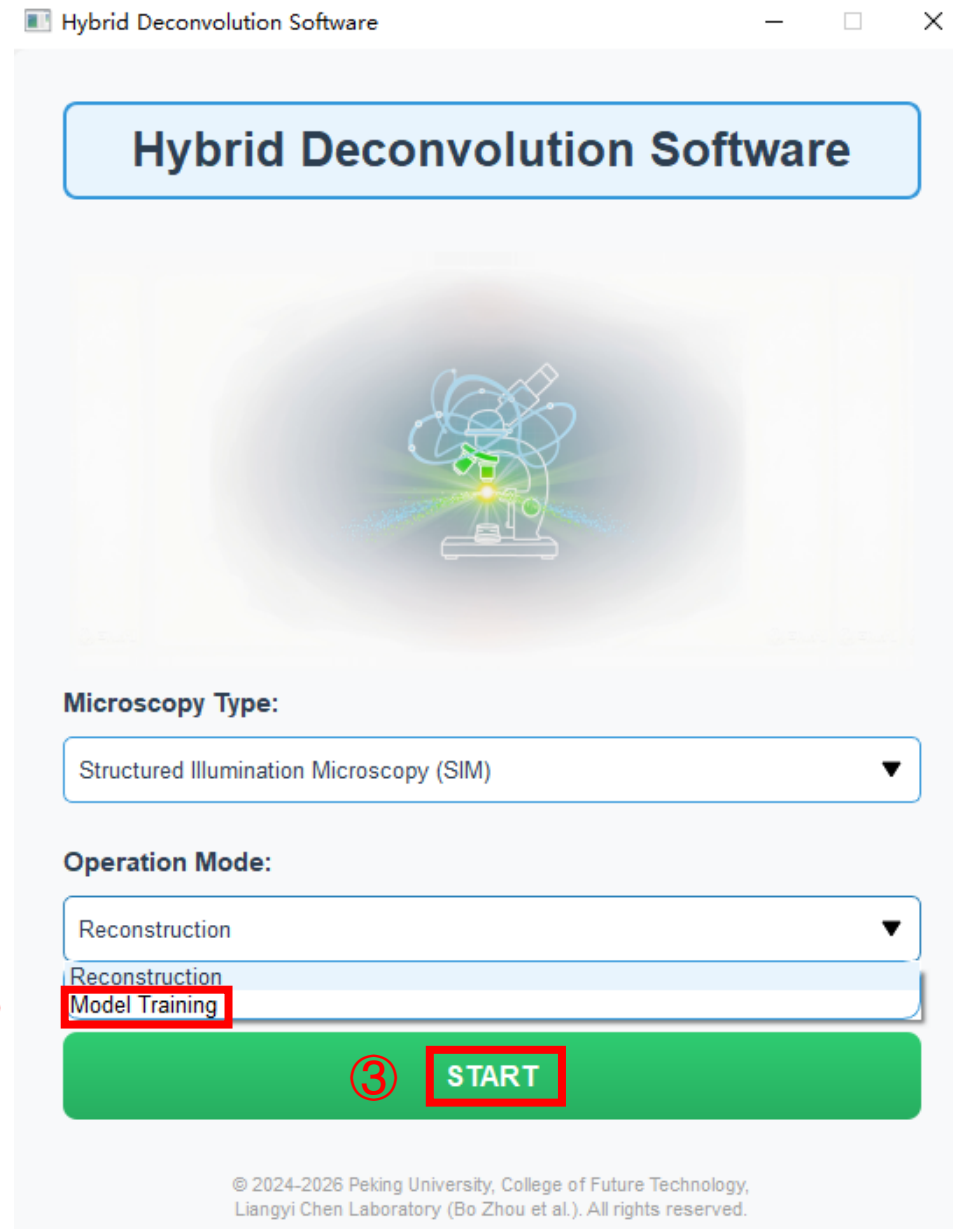
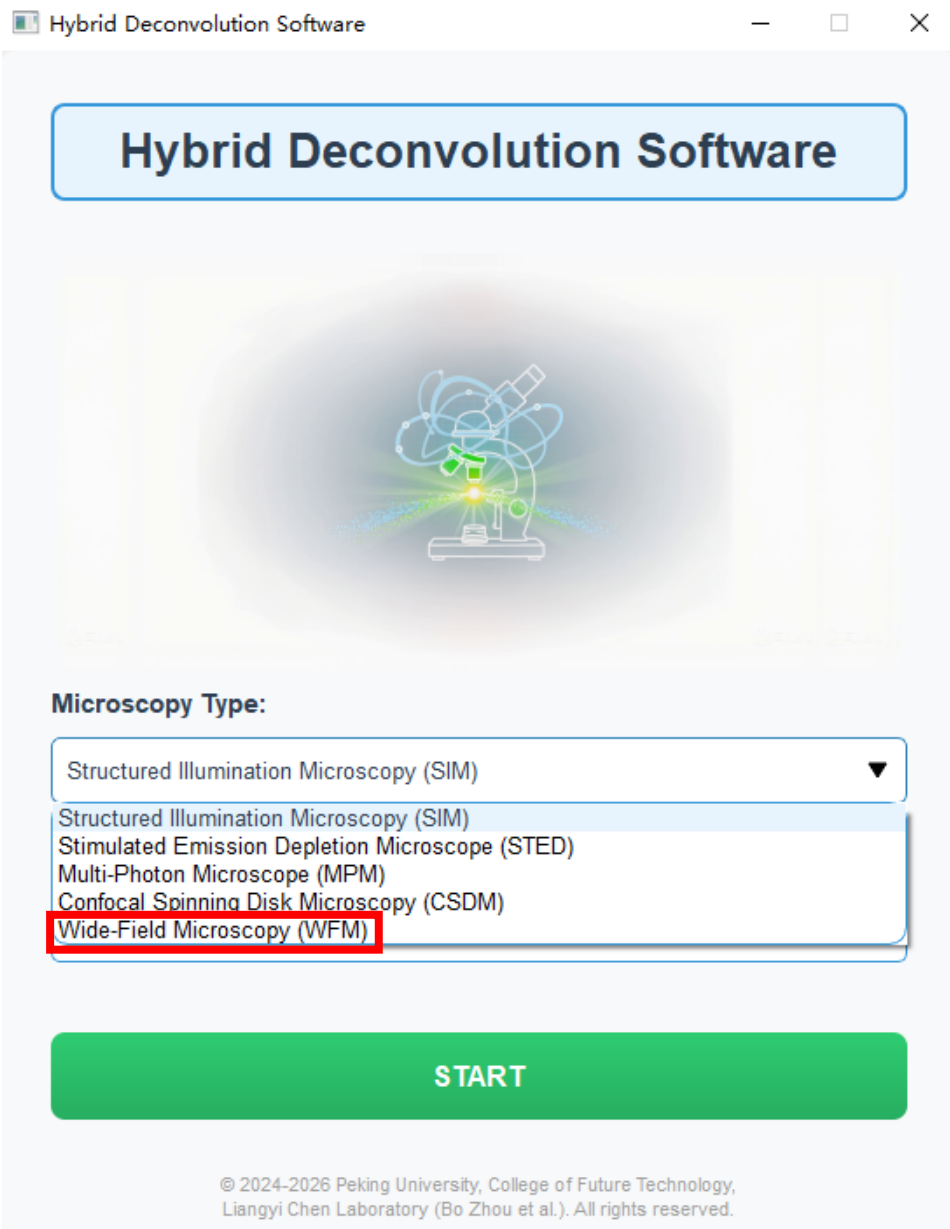
② Run

③ Wait for progress

Tip: The parameters have been covered in the Hybrid-SIM section.

Hybrid-WFM Model Training

Start Hybrid-WFM Model Training GUI



Hybrid-WFM Training GUI layout

① Select parameters

The screenshot displays the 'Hybrid-WFM Training' application window, which is divided into two main panels. The left panel, titled 'Hybrid-WFM Training Parameters', contains two tabs: 'Dataset Generation' and 'TDV-DNN Training'. The 'Dataset Generation' tab is active, showing various parameters for dataset creation, each with a 'Reset' button. At the bottom of this panel are buttons for 'Default Parameters', 'Load Parameters', and 'Export Parameters'. The right panel, titled 'Hybrid-WFM Training Visualization', contains two sub-sections. The top sub-section, 'Training Curves', displays two line graphs: 'Training Loss (x0.1)' vs 'Batch Number' and 'Validation Loss (x0.1)' vs 'Epoch Number'. The bottom sub-section, 'Training Results', shows three side-by-side images: 'TDV-DNN Input', 'TDV-DNN Output', and 'Ground Truth'. Below these images are sliders for 'Display Range' and 'Frame', along with 'Reset Range', 'Reset View', and 'Check Model' buttons. At the bottom of the right panel is a 'Training Log' section showing a list of training events with timestamps, epoch/batch numbers, and loss values. A progress bar at the bottom of the left panel shows the training progress at 10%.

Hybrid-WFM Training Parameters

Dataset Generation | TDV-DNN Training

Dataset Generation Parameters

WFM Raw Data Folder Path: ./Demo_training_WFM [Browse] [Reset]

Emission NA: 1.40 [Reset]

Emission Wavelength (nm): 525.00 [Reset]

Raw Pixel Size (nm): 65.00 [Reset]

Dataset Z Average Number (frame): 20 [Reset]

Dataset Z Skip Number (frame): 5 [Reset]

Dataset XY Block Size (pixel): 256 [Reset]

Dataset XY Block Interval (pixel): 48 [Reset]

Dataset XY Poisson Noise Level: 80.00 [Reset]

Dataset XY Gaussian Noise Level: 80.00 [Reset]

Dataset XY Minimal Heterogeneity: 0.10 [Reset]

Device: cuda:1 (NVIDIA GeForce RTX 3080 Ti, 12.0GB) Detected: 3 device(s)

Operation: Dataset Generation & TDV-DNN Training

[Default Parameters] [Load Parameters] [Export Parameters]

Start Training

Stop Training

10%

Hybrid-WFM Training Visualization

Training Curves

Training Loss (x0.1) vs Batch Number

Validation Loss (x0.1) vs Epoch Number

Training Results

TDV-DNN Input | TDV-DNN Output | Ground Truth

Display Range: [Slider] [Reset Range] [Reset View] [Check Model]

Frame: [Slider] 57/100

Training Log

[11:58:18] Epoch: 0005/0050 | Batch: 0068/0070 | Training-Loss: 0.0003616210 | Time-Interval: 11.73386s
[11:58:29] Epoch: 0005/0050 | Batch: 0069/0070 | Training-Loss: 0.0004052841 | Time-Interval: 11.68205s
[11:58:37] Epoch: 0005/0050 | Batch: 0070/0070 | Training-Loss: 0.0003730508 | Time-Interval: 7.36678s
[11:59:26] Epoch: 0005/0050 | Validation-Loss: 0.0003505269 | Time-Interval: 48.97345s
[11:59:37] Epoch: 0006/0050 | Batch: 0001/0070 | Training-Loss: 0.0003593634 | Time-Interval: 11.73782s
[11:59:49] Epoch: 0006/0050 | Batch: 0002/0070 | Training-Loss: 0.0003981451 | Time-Interval: 11.62355s
[12:00:01] Epoch: 0006/0050 | Batch: 0003/0070 | Training-Loss: 0.0003767489 | Time-Interval: 11.66596s
[12:00:11] Stop TDV-DNN training ...
[12:00:13] TDV-DNN training stopped!
[12:00:13] Ready!

② Run

③ Wait for progress

④ Visualize training process and export TDV-DNN model

Tip: The parameters have been covered in the Hybrid-SIM section.

Thanks!